

# The Prisoner's Dilemma: Theory, Tournaments, Evolution, and Noise

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## Abstract

We study the Prisoner's Dilemma from three angles and test each numerically. Analytically we prove strict dominance and uniqueness of the one-shot equilibrium, backward induction for the finitely repeated game, the discount thresholds for Grim Trigger and Tit-for-Tat in the infinitely repeated game (including the knife-edge on which TFT is subgame perfect), the folk theorems, and the complete phase portrait of the replicator dynamics for ALLC, ALLD and TFT with symbolic Jacobian spectra. Numerically, a purpose-built Python laboratory (pdlab) reproduces every threshold by geometric-stopping simulation, runs an eighteen-strategy Axelrod tournament with and without implementation noise, integrates the replicator equation, simulates the Moran process against exact fixation probabilities, and reproduces the Nowak-May spatial cooperator fraction (0.314 versus the published 0.318). As an original contribution we treat memory-one strategies under implementation error as an exact four-state Markov chain, proving that Tit-for-Tat's self-play payoff drops discontinuously to  $(T+R+P+S)/4$  for any positive error rate while generous strategies lose only linearly, show that selection under noise follows that ranking, and test Press-Dyson extortion against a Q-learning opponent: the learner is pinned to the extortion line but stops short of the full cooperation an idealised adaptive opponent would reach, while extortion itself is driven to extinction in populations. All results regenerate from a single command, and every strategy implementation is cross-checked against an independent library.

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# 1 Introduction

## 1.1 Why the Prisoner’s Dilemma

Two suspects are arrested. Held in separate cells, each is offered the same deal: testify against your accomplice and go free while he serves ten years; stay silent and, if he testifies, you serve the ten years yourself. If both testify, both serve five years; if both stay silent, the prosecutor can only make a minor charge stick and both serve one year. Whatever the other does, testifying is strictly better — and so both testify, and both serve five years, when both could have served one.

This narrative dressing is due to Albert W. Tucker, who used it in 1950 to make an abstract two-by-two game intelligible to a psychology audience at Stanford. The game itself was older by a few months: Merrill Flood and Melvin Dresher had constructed it at the RAND Corporation in January 1950, and had run what is probably the first laboratory experiment on it — a hundred-round repetition between two colleagues, whose behaviour stubbornly refused to match the backward-induction prediction (Flood, 1958). Flood’s report already contains the two observations that

have organised the field ever since: individually rational play destroys jointly available value, and *repetition changes the answer*.

The reason the game has not been retired in the seventy-six years since is that its structure recurs wherever the private cost of a cooperative act is borne by the actor while its benefit accrues to someone else. An arms race is a Prisoner’s Dilemma in which cooperation is disarmament and the temptation is a first-strike advantage. Two firms in a duopoly face one in which cooperation is the monopoly price and the temptation is to undercut. Greenhouse-gas abatement is a many-player version in which every nation prefers that others abate; the free-rider incentive is exactly the dominance argument above. In evolutionary biology, the dilemma is the standard model for the evolution of altruism: a cooperative act that costs the donor  $c$  and delivers  $b > c$  to the recipient is a Prisoner’s Dilemma in which natural selection, acting on individual fitness, favours defection (Axelrod & Hamilton, 1981). That cooperation nonetheless abounds — in cleaner fish, in vampire bats, in human institutions — is the puzzle that the theory in this report exists to resolve.

The resolution offered by game theory is not that the dominance argument is wrong. It is that the one-shot game is the wrong model. Once the same two players expect to meet again, a defection today can be paid for tomorrow, and cooperation can be sustained as an equilibrium of the *repeated* game. Robert Axelrod’s computer tournaments (Axelrod, 1984) gave this abstraction an empirical face: the winner of both tournaments, submitted by Anatol Rapoport, was TIT FOR TAT — cooperate first, then copy the opponent’s last move — a strategy of four lines whose success Axelrod attributed to being nice, retaliatory, forgiving and clear.

## 1.2 Roadmap

Chapter 2 sets up the one-shot game and proves the classical facts: strict dominance, uniqueness of the Nash equilibrium among all mixed profiles, and an exact characterisation of which pure profiles are Pareto-efficient. It also fixes the two parametrisations used throughout: the donation game  $(b, c)$  and the weak dilemma of Nowak and May.

Chapter 3 treats the finitely repeated game. The backward-induction argument is given in full, including the fact — often asserted and rarely proved — that unrelenting defection is the unique subgame-perfect equilibrium *and* the unique on-path Nash outcome for every horizon  $n$ .

Chapter 4 is the analytical core: the infinitely repeated game with discount factor  $\delta$ . We prove that Grim Trigger sustains cooperation exactly when  $\delta \geq (T - R)/(T - P)$ , and derive Axelrod’s collective-stability threshold for TIT FOR TAT by solving a two-state discounted dynamic program, which turns “consider all deviations” into a finite computation. We then show that (TIT FOR TAT, TIT FOR TAT) is subgame perfect only on a knife-edge set of payoffs — never for Axelrod’s  $(5, 3, 1, 0)$  — and close with the Nash and subgame-perfect folk theorems.

Chapter 5 moves from rationality to selection: the replicator equation, evolutionary stability, a complete phase portrait of the three-strategy system  $\{\text{ALLC}, \text{ALLD}, \text{TFT}\}$  with symbolically computed Jacobian spectra, the Moran process and the 1/3 law, and the Nowak–May spatial lattice, whose famous thresholds in  $b$  we derive from explicit neighbourhood score counts.

Later chapters (numerical, empirical) report the computational work; this document contains the theory those chapters rest on.

### 1.3 Contributions

The report has three layers.

**Reproduced theory.** Chapters 2–5 restate the classical results with complete proofs and uniform notation. Where the literature is content with a sketch — the “only two deviations matter” step in Axelrod’s Proposition 2, the failure of subgame perfection for TIT FOR TAT, the lattice thresholds  $b = 9/8, 7/5, 8/5, 9/5$  in the Nowak–May model — we give the argument in full.

**A numerical laboratory.** All quantitative claims are backed by `pdlab`, a typed, tested Python package: `pdlab.game` for the payoff algebra, `pdlab.analytic` for the symbolic thresholds and replicator spectra (computed with `sympy`, not transcribed from the literature), `pdlab.markov` for exact memory-one analysis, and modules for tournaments, evolution, spatial dynamics and reinforcement learning. Every threshold quoted in this report is regenerated from source at build time.

**Original contribution.** Two pieces of work go beyond reproduction. First, an exact treatment of *implementation noise*: with trembling-hand error  $\varepsilon$ , a pair of memory-one strategies  $(p_{CC}, p_{CD}, p_{DC}, p_{DD})$  induces an irreducible Markov chain on the four states  $CC, CD, DC, DD$ , so long-run payoffs are stationary expectations computable in closed form rather than estimated by simulation. This lets us map the robustness of the entire memory-one cube as a function of  $\varepsilon$  — and to show precisely where the simulation-based folklore about noisy tournaments is quantitatively wrong. Second, we pit zero-determinant *extortionate* strategies (Press & Dyson, 2012) against an adaptive  $Q$ -learning opponent (Watkins & Dayan, 1992), testing Press and Dyson’s claim that an evolutionary opponent can be steered: extortion is a Nash-like threat only against an agent whose learning dynamics accept the offered linear relation, and we characterise when the learner instead walks to mutual defection.

## 2 The One-Shot Prisoner’s Dilemma

### 2.1 The normal form

**Definition 2.1 (Finite normal-form game).** A *finite normal-form game* is a triple  $G = \langle N, (A_i)_{i \in N}, (u_i)_{i \in N} \rangle$  in which  $N$  is a finite set of *players*,  $A_i$  is a finite set of *actions* available to player  $i$ , and  $u_i : A \rightarrow \mathbb{R}$  is player  $i$ ’s *payoff function*, defined on the set of *action profiles*  $A = \prod_{j \in N} A_j$ . We write  $a = (a_i, a_{-i})$  to separate player  $i$ ’s component from the others’.

Throughout this report  $N = \{1, 2\}$  and  $A_1 = A_2 = \{C, D\}$ , where  $C$  stands for *cooperate* and  $D$  for *defect*. The game is symmetric:  $u_2(a_1, a_2) = u_1(a_2, a_1)$  for all  $(a_1, a_2)$ , so a single function determines both payoffs.

**Definition 2.2 (Prisoner’s Dilemma).** Fix four real numbers  $T, R, P, S$  (*temptation, reward, punishment, sucker*) and define the symmetric game  $G(T, R, P, S)$  by

$$u_1(C, C) = R, \quad u_1(C, D) = S, \quad u_1(D, C) = T, \quad u_1(D, D) = P,$$

with  $u_2(a_1, a_2) = u_1(a_2, a_1)$ . The game is a (*strict*) *Prisoner’s Dilemma* if

$$T > R > P > S \tag{2.1}$$

and

$$2R > T + S. \tag{2.2}$$

In bimatrix form, with player 1 choosing the row and player 2 the column:

|     | $C$    | $D$    |
|-----|--------|--------|
| $C$ | $R, R$ | $S, T$ |
| $D$ | $T, S$ | $P, P$ |

The canonical numerical instance used throughout this report is Axelrod's  $(T, R, P, S) = (5, 3, 1, 0)$  (Axelrod, 1984), implemented as `PayoffMatrix.axelrod()`:

|     | $C$  | $D$  |
|-----|------|------|
| $C$ | 3, 3 | 0, 5 |
| $D$ | 5, 0 | 1, 1 |

Condition (2.1) is what makes the game a dilemma:  $T > R$  means unilateral defection beats mutual cooperation,  $P > S$  means defection beats cooperation against a defector, and  $R > P$  means mutual cooperation beats mutual defection. Condition (2.2) is a separate requirement, and it is worth being explicit about its role.

**Remark 2.3 (Why  $2R > T + S$ ).** Consider two players who could coordinate on a public randomisation or on an alternating schedule: player 1 plays  $D$  while player 2 plays  $C$  in odd rounds, and the roles swap in even rounds. The long-run average payoff to each is  $(T + S)/2$ . Without (2.2) this is at least  $R$ , and “cooperation” in the sense of  $(C, C)$  would no longer be the jointly best arrangement — the efficient outcome would be *taking turns exploiting each other*, and the entire literature on sustaining  $(C, C)$  by repetition would be pointed at the wrong target. Condition (2.2) says precisely that  $(R, R)$  maximises the sum of payoffs over the feasible set; see Theorem 2.9 and, in the repeated game, Lemma 4.12. Note that (2.2) does not follow from (2.1):  $(T, R, P, S) = (5, 2.4, 1, 0)$  satisfies (2.1) but has  $2R = 4.8 < 5 = T + S$ . The class `PayoffMatrix` enforces (2.2) unconditionally, and (2.1) unless `strict=False`.

## 2.2 Dominance, equilibrium, efficiency

Let  $\Delta(A_i)$  denote the set of probability distributions over  $A_i$ ; an element  $\sigma_i \in \Delta(A_i)$  is a *mixed strategy*, and payoffs are extended to mixed profiles by expectation,

$$u_i(\sigma_1, \sigma_2) = \sum_{a \in A} \sigma_1(a_1) \sigma_2(a_2) u_i(a).$$

**Definition 2.4 (Strict dominance).** Action  $a_i \in A_i$  *strictly dominates*  $a'_i \in A_i$  if  $u_i(a_i, a_{-i}) > u_i(a'_i, a_{-i})$  for every  $a_{-i} \in A_{-i}$ . It *weakly dominates*  $a'_i$  if the inequality holds weakly for every  $a_{-i}$  and strictly for at least one. An action is *strictly (weakly) dominant* if it strictly (weakly) dominates every other action.

**Definition 2.5 (Nash equilibrium).** A mixed profile  $\sigma^* = (\sigma_1^*, \sigma_2^*)$  is a *Nash equilibrium* if for every player  $i$  and every  $\sigma_i \in \Delta(A_i)$ ,  $u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*)$ . It is a *pure equilibrium* if each  $\sigma_i^*$  is degenerate.

**Definition 2.6 (Pareto dominance and efficiency).** For payoff vectors  $v, w \in \mathbb{R}^N$ ,  $w$  *Pareto-dominates*  $v$  if  $w_i \geq v_i$  for all  $i$  with strict inequality for some  $i$ . Given a set  $V \subseteq \mathbb{R}^N$  of attainable

payoff vectors,  $v \in V$  is *Pareto-efficient in  $V$*  if no  $w \in V$  Pareto-dominates it. We take  $V$  to be the *feasible set*

$$V = \text{conv}\{(R, R), (S, T), (T, S), (P, P)\} \subset \mathbb{R}^2, \quad (2.3)$$

the convex hull of the four pure payoff vectors — the payoffs attainable with a public randomising device, and (Chapter 4) the set of long-run average payoffs attainable in the repeated game.

**Theorem 2.7 (Dominance).** In a strict Prisoner's Dilemma,  $D$  strictly dominates  $C$  for both players.

*Proof.* By symmetry it suffices to treat player 1. There are two possible actions of player 2. If  $a_2 = C$ , then  $u_1(D, C) - u_1(C, C) = T - R > 0$  by (2.1). If  $a_2 = D$ , then  $u_1(D, D) - u_1(C, D) = P - S > 0$  by (2.1). Both differences are strictly positive, so  $D$  strictly dominates  $C$ .  $\square$

The two gaps  $(T - R, P - S)$  are returned by `PayoffMatrix.dominance_gap()`; both are strictly positive exactly when  $D$  strictly dominates  $C$ .

**Theorem 2.8 (Uniqueness of equilibrium).** In a strict Prisoner's Dilemma,  $(D, D)$  is the unique Nash equilibrium, in mixed as well as pure strategies.

*Proof.* First,  $(D, D)$  is an equilibrium: by Theorem 2.7,  $u_1(D, D) > u_1(C, D)$ , so no unilateral deviation by player 1 to  $C$  — and hence, by linearity of  $u_1$  in  $\sigma_1$ , no deviation to any mixture — is profitable; symmetrically for player 2.

For uniqueness, let  $\sigma = (\sigma_1, \sigma_2)$  be a Nash equilibrium and suppose  $q := \sigma_1(C) > 0$ . Define  $\sigma'_1$  to be the degenerate strategy on  $D$ . Writing  $r := \sigma_2(C)$ , linearity gives

$$u_1(\sigma'_1, \sigma_2) - u_1(\sigma_1, \sigma_2) = q[u_1(D, \sigma_2) - u_1(C, \sigma_2)] = q[r(T - R) + (1 - r)(P - S)].$$

Since  $r \in [0, 1]$  and both  $T - R$  and  $P - S$  are strictly positive, the bracket is bounded below by  $\min\{T - R, P - S\} > 0$ ; with  $q > 0$  the whole expression is strictly positive. Hence  $\sigma_1$  is not a best response to  $\sigma_2$ , contradicting the equilibrium property. Therefore  $\sigma_1(C) = 0$ , i.e.  $\sigma_1 = D$ , and symmetrically  $\sigma_2 = D$ .  $\square$

Note that the argument uses only strict dominance: in *any* finite game, a strictly dominated action receives probability zero in every Nash equilibrium, and if each player has a strictly dominant action the profile of those actions is the unique equilibrium. This is `analytic.one_shot_facts`, which reports `dd_unique_nash` exactly when `d_dominant` holds.

**Theorem 2.9 (Efficiency classification).** In a strict Prisoner's Dilemma with feasible set  $V$  as in (2.3), the payoff vectors of the profiles  $(C, C)$ ,  $(C, D)$  and  $(D, C)$  are Pareto-efficient in  $V$ , and that of  $(D, D)$  is not. In particular the unique Nash equilibrium is the unique Pareto-inefficient pure profile, and it is Pareto-dominated by  $(C, C)$ .

*Proof.* (i)  $(D, D)$  is inefficient. Its payoff vector is  $(P, P)$  and  $(R, R) \in V$  with  $R > P$  in both coordinates, so  $(R, R)$  Pareto-dominates  $(P, P)$ .

(ii)  $(C, D)$  is efficient. Its payoff vector is  $(S, T)$ . By (2.1),  $T = \max\{T, R, P, S\}$ , so every vertex of  $V$  has second coordinate at most  $T$ ; since the second coordinate is a linear functional, its maximum over the polytope  $V$  equals its maximum over the vertices, namely  $T$ , and among the vertices it is attained only at  $(S, T)$ . Suppose  $w \in V$  Pareto-dominates  $(S, T)$ ; then  $w_2 \geq T$ , hence  $w_2 = T$ , hence  $w$  lies on the face  $\{v \in V : v_2 = T\}$ . A face of a polytope is the convex hull of the vertices it contains, and the only such vertex is  $(S, T)$ ; therefore  $w = (S, T)$ , which does not Pareto-dominate itself. Hence  $(S, T)$  is efficient. The case of  $(D, C)$  is symmetric.

(iii)  $(C, C)$  is efficient. Consider the linear functional  $\Sigma(v) = v_1 + v_2$ . Its values at the four vertices are  $2R$ ,  $T + S$ ,  $T + S$  and  $2P$ . By (2.2),  $2R > T + S$ , and by (2.1),  $R > P$  so  $2R > 2P$ . Hence  $\max_{v \in V} \Sigma(v) = 2R$ , attained only at  $(R, R)$ . If  $w \in V$  Pareto-dominated  $(R, R)$  then  $\Sigma(w) > \Sigma(R, R) = 2R$ , a contradiction.  $\square$

The classification is worth emphasising because it is frequently misstated. The Prisoner’s Dilemma is *not* a game in which the two “exploitative” outcomes are socially bad; they are Pareto-efficient, because the exploiter cannot be made better off at all. What is bad about them is that they are unfair and that they are not sustainable as equilibria. The genuinely defective outcome is  $(D, D)$ , and it is the one the players are driven to.

## 2.3 Two reparametrisations

### 2.3.1 The donation game

**Definition 2.10 (Donation game).** Fix  $b > c > 0$ . In the *donation game*  $\text{Don}(b, c)$ , a cooperator pays a cost  $c$  to confer a benefit  $b$  on the co-player; a defector does nothing. The induced payoffs are

$$T = b, \quad R = b - c, \quad P = 0, \quad S = -c.$$

**Proposition 2.11.**  $\text{Don}(b, c)$  is a strict Prisoner’s Dilemma for every  $b > c > 0$ .

*Proof.* Order:  $T - R = c > 0$ ,  $R - P = b - c > 0$ ,  $P - S = c > 0$ , so  $T > R > P > S$ . Additivity:  $2R - (T + S) = 2(b - c) - (b - c) = b - c > 0$ .  $\square$

The donation game is the standard vehicle for the biological literature, because it has one degree of freedom after normalisation: setting  $c = 1$  leaves the *benefit-to-cost ratio*  $b/c$  as the only parameter, and the classical conditions for the evolution of cooperation are all of the form  $b/c > \text{something}$  (Chapter 5). It is implemented as `PayoffMatrix.donation(b, c)`.

### 2.3.2 The weak dilemma of Nowak and May

**Definition 2.12 (Weak dilemma).** For  $1 < b < 2$ , the *Nowak–May* parametrisation (Nowak & May, 1992) is

$$R = 1, \quad T = b, \quad P = S = 0,$$

a distinct degenerate parametrisation in which the sucker payoff is raised to the punishment payoff (it is not a limit of the donation game: rescaling  $\text{Don}(b, c)$  to  $R = 1$  and letting  $c \rightarrow 0$  gives the trivial game  $T = R = 1$ ). Its payoff bimatrix is

|     | $C$     | $D$    |
|-----|---------|--------|
| $C$ | 1, 1    | 0, $b$ |
| $D$ | $b$ , 0 | 0, 0   |

Here  $T > R \geq P \geq S$  holds with  $P = S$ , so (2.1) fails and only the weak ordering survives; (2.2) becomes  $2 > b$ , which is exactly the stated restriction  $b < 2$ . `PayoffMatrix.nowak_may(b)` enforces  $1 < b < 2$  and is constructed with `strict=False`.

**Theorem 2.13 (Equilibrium structure of the weak dilemma).** Let  $1 < b < 2$ . Then:

1.  $D$  weakly — but not strictly — dominates  $C$  for both players.

2.  $(D, D)$  is a Nash equilibrium, but it is *not* the unique one: the set of Nash equilibria is

$$\mathcal{N} = \{(\sigma_1, D) : \sigma_1 \in \Delta(\{C, D\})\} \cup \{(D, \sigma_2) : \sigma_2 \in \Delta(\{C, D\})\}.$$

3.  $(D, D)$  is the unique Nash equilibrium surviving the elimination of weakly dominated strategies, and the unique trembling-hand perfect equilibrium.

*Proof.* (1) Against  $C$ :  $u_1(D, C) - u_1(C, C) = b - 1 > 0$ . Against  $D$ :  $u_1(D, D) - u_1(C, D) = P - S = 0$ . So the dominance is weak and not strict.

- (2) Write  $q = \sigma_1(C)$ ,  $r = \sigma_2(C)$ . Player 1's payoffs are  $u_1(C, \sigma_2) = r$  and  $u_1(D, \sigma_2) = rb$ , so  $D$  is a best response for every  $r$ , and  $C$  is a best response iff  $r = 0$ , i.e. iff  $\sigma_2 = D$ . Symmetrically for player 2. Consequently  $(\sigma_1, \sigma_2)$  is an equilibrium iff  $[\sigma_1 = D \text{ or } \sigma_2 = D]$  and  $[\sigma_2 = D \text{ or } \sigma_1 = D]$ , i.e. iff at least one player plays  $D$  with probability one. This is exactly  $\mathcal{N}$ .
- (3)  $C$  is weakly dominated for both players, so a single round of elimination leaves the profile  $(D, D)$ . For perfection, let  $(\sigma_1^\eta, \sigma_2^\eta)$  be any sequence of completely mixed profiles converging to a candidate equilibrium, and suppose  $\sigma_1(C) > 0$  in the limit. Along the sequence  $r^\eta := \sigma_2^\eta(C) > 0$ , so  $u_1(D, \sigma_2^\eta) - u_1(C, \sigma_2^\eta) = r^\eta(b - 1) > 0$  and  $C$  is never a best response to  $\sigma_2^\eta$ ; hence in the limit player 1 must put probability zero on  $C$ . The same holds for player 2, so  $(D, D)$  is the only trembling-hand perfect equilibrium; and it is perfect, being a limit of best responses to any perturbation by the same computation.  $\square$

**Remark 2.14.** Part (2) is a genuine difference from the strict game and a standard trap. Because  $P = S$ , a player facing a certain defector is *indifferent* between  $C$  and  $D$ , so profiles such as  $(C, D)$  — one player is exploited and does not mind, because being exploited costs nothing in this parametrisation — satisfy the equilibrium conditions. The weak dilemma is used in this report only in the spatial model of Section 5.5, where the dynamics is imitation-based rather than equilibrium-based and this multiplicity is immaterial; wherever equilibrium reasoning is done we use the strict game.

## 3 The Finitely Repeated Game

### 3.1 Definitions

Fix a stage game  $G = \langle \{1, 2\}, \{C, D\}^2, (u_1, u_2) \rangle$  satisfying (2.1)–(2.2), and an integer horizon  $n \geq 1$ .

**Definition 3.1 (The  $n$ -fold repeated game).** The  $n$ -fold repetition  $G^n$  is the extensive game with perfect recall in which  $G$  is played in rounds  $t = 1, \dots, n$ , both players observing the realised action profile at the end of each round (*perfect monitoring*). A *history of length  $t$*  is a sequence  $h^t = (a^1, \dots, a^t) \in A^t$  with  $A = \{C, D\}^2$ ;  $h^0 = \emptyset$  is the empty history and  $H = \bigcup_{t=0}^{n-1} A^t$  is the set of histories at which play continues. A (behaviour) *strategy* for player  $i$  is a map

$$s_i : H \longrightarrow \Delta(\{C, D\}),$$

assigning to each history a — possibly randomised — action;  $s_i$  is *pure* if every  $s_i(h)$  is degenerate. Payoffs in  $G^n$  are the undiscounted sums

$$U_i^n(s_1, s_2) = \mathbb{E} \left[ \sum_{t=1}^n u_i(a^t) \right],$$

the expectation taken over the play path induced by  $(s_1, s_2)$ .

**Definition 3.2 (Subgame; SPE).** For each  $h \in H$  of length  $t$ , the *subgame*  $G^n|_h$  is the  $(n-t)$ -fold repetition of  $G$  played after  $h$ ; the *continuation strategy*  $s_i|_h$  is defined by  $s_i|_h(h') = s_i(hh')$ , and the continuation payoff is  $U_i^{n-t}(s_1|_h, s_2|_h)$ . A profile  $(s_1^*, s_2^*)$  is a *subgame-perfect equilibrium* (SPE) of  $G^n$  if for every  $h \in H$  the profile  $(s_1^*|_h, s_2^*|_h)$  is a Nash equilibrium of  $G^n|_h$ . Every SPE is a Nash equilibrium (take  $h = \emptyset$ ); the converse fails.

Because  $G^n$  has perfect recall and finitely many nodes, subgame perfection can be checked one round at a time.

**Theorem 3.3 (One-shot deviation principle, finite horizon).** A strategy profile  $s^* = (s_1^*, s_2^*)$  of  $G^n$  is a subgame-perfect equilibrium if and only if it admits no profitable *one-shot deviation*: there is no player  $i$ , no history  $h \in H$ , and no action  $a_i \in \{C, D\}$  such that player  $i$  strictly increases her continuation payoff in  $G^n|_h$  by playing  $a_i$  at  $h$  and following  $s_i^*$  at every other history.

*Proof (sketch; see (Fudenberg & Tirole, 1991, Thm. 4.1) or (Osborne & Rubinstein, 1994, Lemma 98.2)).* Necessity is immediate: a one-shot deviation is a deviation. For sufficiency, suppose  $s^*$  admits no profitable one-shot deviation but some  $\hat{s}_i$  strictly improves player  $i$ 's payoff in the subgame after some  $h$ , by an amount  $\eta > 0$ . Since  $G^n|_h$  has finite depth,  $\hat{s}_i$  differs from  $s_i^*$  at only finitely many histories; let  $\tau$  be the *last* round at which they differ. Replacing  $\hat{s}_i$  by  $s_i^*$  from round  $\tau$  onwards cannot lower player  $i$ 's payoff, since at every round- $\tau$  history the modification is a one-shot deviation from  $s_i^*$  in the continuation, and by hypothesis such deviations are not strictly profitable. Iterating backwards over the finitely many rounds at which they differ produces  $s_i^*$  itself with payoff at least that of  $\hat{s}_i$ , contradicting  $\eta > 0$ .  $\square$

Write ALLD for the strategy  $s_i^D(h) = D$  for every  $h \in H$ : unconditional defection.

### 3.2 Backward induction

**Theorem 3.4 (Unravelling; uniqueness of the SPE).** For every  $n \geq 1$ , the  $n$ -fold repeated Prisoner's Dilemma  $G^n$  has exactly one subgame-perfect equilibrium, namely (ALLD, ALLD). Its payoff is  $(nP, nP)$ , and the same is true in every subgame.

*Proof.* We prove by induction on  $k$  the statement  $\mathcal{P}(k)$ : *in every subgame of  $G^n$  with  $k$  rounds remaining, the unique SPE is unconditional defection by both players, with continuation payoff  $(kP, kP)$ .*

$\mathcal{P}(1)$ : a one-round subgame is a copy of the stage game  $G$ , whose unique Nash equilibrium is  $(D, D)$  with payoff  $(P, P)$  by Theorem 2.8.

$\mathcal{P}(k-1) \Rightarrow \mathcal{P}(k)$  for  $k \geq 2$ : let  $h$  be a history with  $k$  rounds remaining and let  $s^*$  be an SPE of  $G^n$ . Every history  $ha$ ,  $a \in A$ , has  $k-1$  rounds remaining, so by  $\mathcal{P}(k-1)$  the continuation profile  $s^*|_{ha}$  is unconditional defection and yields  $((k-1)P, (k-1)P)$  — *for every  $a \in A$ , on path and off*. Hence the continuation payoff after the current round is the constant  $(k-1)P$  for each player, independent of the current round's actions. Player  $i$ 's payoff in  $G^n|_h$  is therefore

$$u_i(\sigma_1, \sigma_2) + (k-1)P,$$

where  $\sigma_j = s_j^*(h)$ . Adding a constant does not change the best-response correspondence, so  $(\sigma_1, \sigma_2)$  is a Nash equilibrium of  $G^n|_h$  if and only if it is a Nash equilibrium of the stage game  $G$ ; by Theorem 2.8 this forces  $\sigma_1 = \sigma_2 = D$ , and the continuation payoff is  $P + (k-1)P = kP$ . This establishes  $\mathcal{P}(k)$ .

Applying  $\mathcal{P}(k)$  at every history — including  $h = \emptyset$  with  $k = n$  — shows that  $s^*$  prescribes  $D$  everywhere, i.e.  $s^* = (\text{ALLD}, \text{ALLD})$ ; and this profile is an SPE because at every history it induces the stage equilibrium with history-independent continuations, so Theorem 3.3 applies with all one-shot deviation gains equal to  $u_i(a_i, D) - u_i(D, D) \leq 0$ .  $\square$

The engine of the proof is the phrase *independent of the current round's actions*: because the continuation is pinned down uniquely and identically after every action profile, no threat and no promise can be attached to today's behaviour, and the game collapses to  $n$  independent copies of  $G$ .

### 3.3 Nash equilibria without perfection

Subgame perfection is a strong requirement, and one might hope that Nash equilibrium alone permits cooperation, sustained by threats that are incredible but never tested. It does not.

**Theorem 3.5 (Uniqueness of the equilibrium path).** Let  $s^* = (s_1^*, s_2^*)$  be *any* Nash equilibrium of  $G^n$ . Then the play path induced by  $s^*$  is  $(D, D)$  in every round, with probability one, and  $U_i^n(s^*) = nP$ .

*Proof.* Note first that each player can *guarantee*  $P$  per round:  $\min_{a_j} u_i(D, a_j) = \min\{T, P\} = P$ , so by playing  $D$  in every remaining round a player secures at least  $mP$  in any  $m$ -round continuation, whatever the opponent does. (This is the minmax bound of Definition 4.13.)

We show by downward induction on  $t = n, n-1, \dots, 1$  that the on-path round- $t$  profile is  $(D, D)$  almost surely. Consider the statement  $\mathcal{Q}(t)$ : *on every history reached with positive probability under  $s^*$ , rounds  $t, t+1, \dots, n$  are played as  $(D, D)$ .*

$\mathcal{Q}(n+1)$  is vacuously true. Assume  $\mathcal{Q}(t+1)$ ; let  $h$  be a positive-probability history of length  $t-1$ , and let  $\sigma_j = s_j^*(h)$ . By  $\mathcal{Q}(t+1)$ , the continuation payoff from round  $t+1$  onwards, *along the equilibrium path from  $h$* , equals  $(n-t)P$  for each player. Hence

$$U_i^{n-t+1}(s^*|_h) = u_i(\sigma_1, \sigma_2) + (n-t)P.$$

Now fix  $i$  and let  $b_i \in \{C, D\}$  be a best response in the stage game to  $\sigma_{-i}$ . Consider the deviation  $\hat{s}_i$ : follow  $s_i^*$  until  $h$ ; at  $h$  play  $b_i$ ; from round  $t+1$  onwards play  $D$  unconditionally. By the guarantee above,

$$U_i^{n-t+1}(\hat{s}_i, s_{-i}^*|_h) \geq u_i(b_i, \sigma_{-i}) + (n-t)P.$$

Since  $h$  has positive probability and  $s^*$  is a Nash equilibrium, deviating in this way cannot be strictly profitable in the whole game, hence not in the continuation from  $h$ :

$$u_i(\sigma_1, \sigma_2) + (n-t)P \geq u_i(b_i, \sigma_{-i}) + (n-t)P \implies u_i(\sigma_i, \sigma_{-i}) \geq \max_{b_i} u_i(b_i, \sigma_{-i}).$$

So  $\sigma_i$  is a stage-game best response to  $\sigma_{-i}$  for each  $i$ ; that is,  $(\sigma_1, \sigma_2)$  is a Nash equilibrium of  $G$ , and Theorem 2.8 gives  $\sigma_1 = \sigma_2 = D$ . This proves  $\mathcal{Q}(t)$ , and the induction down to  $t = 1$  gives the claim.  $\square$

Theorem 3.5 does not say that  $(\text{ALLD}, \text{ALLD})$  is the unique Nash equilibrium of  $G^n$  — it is not, for  $n \geq 2$ . Off-path behaviour is unconstrained: for  $n = 2$ , the profile “play  $D$  in round 1; in round 2 play  $D$  if round 1 was  $(D, D)$  and play  $C$  otherwise” is a Nash equilibrium (the round-2 prescription is never invoked on path), yet it is not subgame perfect. What Theorem 3.5 does say is that the *outcome* is invariant: no Nash equilibrium of the finitely repeated Prisoner's Dilemma produces a single cooperative round.

**Remark 3.6 (Contrast: multiple stage equilibria).** Both theorems lean on the *uniqueness* of the stage equilibrium. Benoit and Krishna (Benoit & Krishna, 1985) show that when the stage game has multiple Nash equilibria with distinct payoffs for each player, the finitely repeated game admits a folk theorem: as  $n \rightarrow \infty$ , essentially every feasible and individually rational payoff can be supported in subgame-perfect equilibrium, the terminal rounds being used to reward or punish with different stage equilibria. The Prisoner’s Dilemma is exactly the case where this device is unavailable, which is why finite repetition buys nothing and infinite repetition (Chapter 4) buys everything.

**Remark 3.7 (The end-game effect and bounded rationality).** Human subjects do not unravel. In finitely repeated Prisoner’s Dilemmas and their relatives, cooperation is typically high in early rounds and collapses only in the last few; Selten and Stoecker (Selten & Stoecker, 1986) documented, in an experiment of 25 successive ten-round supergames with changing anonymous partners, that subjects learn to defect progressively *earlier* across supergames, but that the learning stabilises well short of round one, exactly as a bounded-rationality model of backward induction with a limited number of steps predicts. The leading rational-choice explanation is reputational: Kreps, Milgrom, Roberts and Wilson (Kreps et al., 1982) show that if each player attaches even a small prior probability to the opponent being a behavioural type committed to TIT FOR TAT, then in the resulting game of incomplete information there are sequential equilibria in which both players cooperate in all but the last few rounds, with the number of defecting rounds bounded independently of  $n$ . A vanishing amount of incomplete information thus overturns Theorem 3.4 completely — a warning that the unravelling result is fragile in exactly the direction the data point.

## 4 The Infinitely Repeated Game

### 4.1 Discounting and strategies

**Definition 4.1 (The discounted supergame).** Fix a stage Prisoner’s Dilemma  $G$  and a *discount factor*  $\delta \in [0, 1)$ . The supergame  $G^\infty(\delta)$  is played in rounds  $t = 0, 1, 2, \dots$  with perfect monitoring. Histories, subgames and (behaviour) strategies are defined as in Definitions 3.1–3.2, with  $H = \bigcup_{t \geq 0} A^t$  now infinite. Given a strategy profile  $s = (s_1, s_2)$  inducing the (random) action path  $(a^t)_{t \geq 0}$ , define the *unnormalised* and *normalised* discounted payoffs

$$\widetilde{U}_i(s) = \mathbb{E} \left[ \sum_{t=0}^{\infty} \delta^t u_i(a^t) \right], \quad U_i(s) = (1 - \delta) \widetilde{U}_i(s). \quad (4.1)$$

Both series converge absolutely because stage payoffs are bounded by  $\max\{|T|, |S|\}$  and  $\delta < 1$ .

The two conventions differ by the positive constant  $1 - \delta$  and therefore induce identical preferences, best responses and equilibria. The normalised version has the advantage that a constant stream  $\pi$  is worth exactly  $\pi$ , which makes payoffs comparable across  $\delta$  and puts them in the same units as the stage game; we use it whenever payoffs of different strategy pairs are compared (Chapter 5 and all payoff matrices), and the unnormalised version inside deviation calculations, where expressions such as  $R/(1 - \delta)$  are cleaner. Every threshold in this chapter is invariant to the choice.

**Remark 4.2 (Two readings of  $\delta$ ).** Algebraically,  $\delta$  is a discount factor: a payoff received one round later is worth  $\delta$  times as much. Behaviourally, it is a *continuation probability*: if after each round the interaction ends with probability  $1 - \delta$  and continues with probability  $\delta$ , then the expected number of rounds in which round  $t$ ’s payoff is actually collected is  $\delta^t$ , and expected total payoff is exactly  $\widetilde{U}_i$ . The expected length of the interaction is  $\sum_{t \geq 0} \delta^t = 1/(1 - \delta)$ . Under this

reading,  $\delta$  measures “the shadow of the future” (Axelrod, 1984) — how likely the players are to meet again — and thresholds of the form  $\delta \geq \delta^*$  read as *cooperation requires a sufficiently long expected relationship*.

**Theorem 4.3 (One-shot deviation principle, infinite horizon).** Let  $\delta \in [0, 1)$ . A strategy profile  $s^*$  of  $G^\infty(\delta)$  is a subgame-perfect equilibrium if and only if no player has a profitable one-shot deviation at any history  $h \in H$ .

*Proof (sketch; (Fudenberg & Tirole, 1991, Thm. 4.2), (Osborne & Rubinstein, 1994, Lemma 98.2)).* Necessity is trivial. For sufficiency the finite-horizon argument of Theorem 3.3 is supplemented by *continuity at infinity*: the payoff function (4.1) satisfies

$$\sup_{s, s': s \text{ and } s' \text{ agree in rounds } 0, \dots, \tau-1} |\widetilde{U}_i(s) - \widetilde{U}_i(s')| \leq \frac{\delta^\tau}{1-\delta} (\max_a u_i(a) - \min_a u_i(a)) \xrightarrow{\tau \rightarrow \infty} 0,$$

because stage payoffs are bounded and  $\delta < 1$ . Suppose  $\hat{s}_i$  improves on  $s_i^*$  in the subgame after  $h$  by  $\eta > 0$ . Choose  $\tau$  with the bound above smaller than  $\eta$ , and let  $\hat{s}_i^\tau$  agree with  $\hat{s}_i$  for the first  $\tau$  rounds after  $h$  and with  $s_i^*$  thereafter; then  $\hat{s}_i^\tau$  still improves on  $s_i^*$  by at least  $\eta$  minus that bound, hence strictly. But  $\hat{s}_i^\tau$  differs from  $s_i^*$  at finitely many rounds, so the backward-unwinding argument of Theorem 3.3 applies verbatim and produces a profitable one-shot deviation, a contradiction.  $\square$

Without continuity at infinity the principle is false; it is the discounting, not the repetition, that makes local checking sufficient.

## 4.2 Grim Trigger

**Definition 4.4 (Grim Trigger).** The strategy GRIM plays  $C$  at the empty history and at every history in which the realised profile was  $(C, C)$  in every past round; it plays  $D$  at every other history. Equivalently: cooperate until someone defects, then defect forever.

GRIM partitions histories into two *phases*: the *cooperative phase*, consisting of histories in which  $(C, C)$  occurred in all past rounds, and the *punishment phase*, all other histories. The phase is common knowledge under perfect monitoring, and once entered the punishment phase is absorbing.

**Theorem 4.5 (Grim Trigger).** (GRIM, GRIM) is a subgame-perfect equilibrium of  $G^\infty(\delta)$  if and only if

$$\delta \geq \delta^G := \frac{T - R}{T - P}. \quad (4.2)$$

For Axelrod’s  $(T, R, P, S) = (5, 3, 1, 0)$ ,  $\delta^G = 2/4 = 1/2$ .

*Proof.* By Theorem 4.3 it suffices to check one-shot deviations at every history; by symmetry we check them for player 1, and by the phase structure there are exactly two cases.

*Cooperative phase.* Let  $h$  be a history in the cooperative phase. Conforming, both players cooperate forever, giving

$$\widetilde{U}_1(\text{conform}) = \sum_{t \geq 0} \delta^t R = \frac{R}{1-\delta}.$$

The one-shot deviation is to play  $D$  now and revert to GRIM afterwards. This yields  $T$  this round; the resulting history lies in the punishment phase for both players, so from the next round both play  $D$  forever, giving  $P$  per round:

$$\widetilde{U}_1(\text{deviate}) = T + \sum_{t \geq 1} \delta^t P = T + \frac{\delta P}{1-\delta}.$$

No profitable deviation therefore requires

$$\frac{R}{1-\delta} \geq T + \frac{\delta P}{1-\delta} \iff R \geq (1-\delta)T + \delta P \iff \delta(T-P) \geq T-R \iff \delta \geq \frac{T-R}{T-P},$$

using  $T > P$  so that the division preserves the inequality. Deviating to  $C$  is not a deviation at all, since GRIM prescribes  $C$  here.

*Punishment phase.* Let  $h$  be a history in the punishment phase. Both continuation strategies prescribe  $D$  unconditionally — the punishment phase is absorbing, so no future history leaves it — and the continuation payoff is  $P/(1-\delta)$ . The one-shot deviation to  $C$  yields  $S$  this round, after which both again play  $D$  forever:

$$S + \frac{\delta P}{1-\delta} < P + \frac{\delta P}{1-\delta} = \frac{P}{1-\delta},$$

strictly, because  $S < P$ . So the deviation is strictly unprofitable, for every  $\delta \in [0, 1)$ ; no condition arises. (This is the substantive content of subgame perfection here: the threat is credible because carrying it out is itself an equilibrium, ALLD being a stage best reply to ALLD.)

Both cases are covered, so by Theorem 4.3 the profile is an SPE precisely when (4.2) holds; and if (4.2) fails the cooperative-phase deviation is strictly profitable, so the profile is not even a Nash equilibrium.  $\square$

The threshold is computed symbolically by `analytic.grim_threshold_symbolic()`, which returns  $(T-R)/(T-P)$ .

### 4.3 Tit for Tat

**Definition 4.6 (Tit for Tat).** The strategy TFT plays  $C$  at the empty history, and at a history  $h^t = (a^0, \dots, a^{t-1})$  plays the opponent's previous action  $a_{-i}^{t-1}$ .

TFT is *nice* (never the first to defect), *retaliatory*, *forgiving* (a single cooperation restores cooperation) and *clear*. It won both of Axelrod's tournaments (Axelrod, 1984). The following is Axelrod's Proposition 2 (his "collective stability" of TFT), stated as a Nash-equilibrium condition. The difficulty in proving it honestly is the phrase *all deviations*: the deviating player has a continuum of behaviour strategies. The resolution is that against TFT the problem is a two-state Markov decision process.

**Lemma 4.7 (The best reply to TFT).** Fix  $\delta \in [0, 1)$  and suppose player 2 plays TFT. Then player 1's optimisation problem is a discounted Markov decision process with state space  $\Theta = \{\theta_C, \theta_D\}$  —  $\theta_x$  meaning "the opponent plays  $x$  this round" — action space  $\{C, D\}$ , deterministic transition  $\theta \mapsto \theta_a$  upon choosing  $a$ , reward  $u_1(a, x)$  in state  $\theta_x$ , and initial state  $\theta_C$ . Its optimal values  $V_C, V_D$  (unnormalised) satisfy the Bellman equations

$$V_C = \max\{R + \delta V_C, T + \delta V_D\}, \quad V_D = \max\{S + \delta V_C, P + \delta V_D\}, \quad (4.3)$$

and

$$V_C = \max\left\{ \underbrace{\frac{R}{1-\delta}}_{\text{always } C}, \underbrace{T + \frac{\delta P}{1-\delta}}_{\text{always } D}, \underbrace{\frac{T+\delta S}{1-\delta^2}}_{\text{alternate } D, C, D, C, \dots} \right\}. \quad (4.4)$$

*Proof.* TFT's action in round  $t \geq 1$  is player 1's action in round  $t-1$ , and its action in round 0 is  $C$ . Hence the opponent's current action is a deterministic function of player 1's own previous action;

taking that action as the state makes player 1's stage reward and transition depend on the current state and current action only, which is precisely a stationary MDP with the stated data. It has two states, two actions, bounded rewards and  $\delta < 1$ , so by standard discounted dynamic programming the Bellman operator is a  $\delta$ -contraction on  $\mathbb{R}^\Theta$ ,  $(V_C, V_D)$  is its unique fixed point, it equals the supremum of  $\widetilde{U}_1$  over *all* strategies of player 1 (history-dependent and randomised included), and the supremum is attained by a deterministic stationary policy.

There are exactly four deterministic stationary policies  $\pi : \Theta \rightarrow \{C, D\}$ , and since transitions are deterministic each generates an eventually periodic payoff stream from  $\theta_C$ :

- $\pi(\theta_C) = C$ ,  $\pi(\theta_D) = C$  (unconditional  $C$ ): the state never leaves  $\theta_C$ ; stream  $R, R, R, \dots$ ; value  $R/(1 - \delta)$ .
- $\pi(\theta_C) = C$ ,  $\pi(\theta_D) = D$  (TFT itself): starting at  $\theta_C$  the state never leaves  $\theta_C$ ; stream  $R, R, R, \dots$ ; value  $R/(1 - \delta)$ .
- $\pi(\theta_C) = D$ ,  $\pi(\theta_D) = D$  (unconditional  $D$ ): stream  $T, P, P, \dots$ ; value  $T + \delta P/(1 - \delta)$ .
- $\pi(\theta_C) = D$ ,  $\pi(\theta_D) = C$  (ALT, “defect, then cooperate, then defect, ...”): the state alternates  $\theta_C, \theta_D, \theta_C, \dots$ ; stream  $T, S, T, S, \dots$ ; value  $\sum_{k \geq 0} \delta^{2k} (T + \delta S) = (T + \delta S)/(1 - \delta^2)$ .

Taking the maximum over the four values — two of which coincide — yields (4.4).  $\square$

Lemma 4.7 is the precise sense in which “only two deviations matter”: the *only* candidate improvements on permanent cooperation are permanent defection and the alternating exploitation cycle.

**Theorem 4.8 (Axelrod's Proposition 2).** (TFT, TFT) is a Nash equilibrium of  $G^\infty(\delta)$  if and only if

$$\delta \geq \delta^{\text{TFT}} := \max \left\{ \frac{T - R}{T - P}, \frac{T - R}{R - S} \right\}. \quad (4.5)$$

For  $(5, 3, 1, 0)$ :  $\max\{2/4, 2/3\} = 2/3$ .

*Proof.* (TFT, TFT) produces mutual cooperation in every round, so each player's equilibrium payoff is  $R/(1 - \delta)$ . By symmetry, the profile is a Nash equilibrium iff no strategy of player 1 earns more than  $R/(1 - \delta)$  against TFT, i.e. iff  $V_C = R/(1 - \delta)$  in Lemma 4.7. By (4.4) this holds iff  $R/(1 - \delta)$  dominates the other two candidates.

Against unconditional  $D$ :

$$\frac{R}{1 - \delta} \geq T + \frac{\delta P}{1 - \delta} \iff R \geq (1 - \delta)T + \delta P \iff \delta \geq \frac{T - R}{T - P}.$$

Against ALT: multiplying by  $(1 - \delta^2) = (1 - \delta)(1 + \delta) > 0$ ,

$$\frac{R}{1 - \delta} \geq \frac{T + \delta S}{1 - \delta^2} \iff R(1 + \delta) \geq T + \delta S \iff \delta(R - S) \geq T - R \iff \delta \geq \frac{T - R}{R - S},$$

using  $R > S$ . The conjunction of the two conditions is (4.5).  $\square$

`analytic.tft_nash_threshold_symbolic()` returns exactly  $\max\{(T - R)/(T - P), (T - R)/(R - S)\}$ . Note that neither term dominates the other in general: the second exceeds the first iff  $R - S < T - P$ , i.e. iff  $R + P < T + S$ , which holds for  $(5, 3, 1, 0)$  ( $4 < 5$ ) but fails for, say,  $(5, 4, 3, 0)$ .

#### 4.4 Tit for Tat is not subgame perfect

Nash equilibrium tolerates incredible threats. TFT contains one.

Under (TFT, TFT) the continuation play after a history depends only on the last realised profile, so there are four continuation regimes. Write them from player 1's point of view, with  $\tilde{V}(h)$  her unnormalised continuation payoff and  $\tilde{V}^{\text{dev}}(h)$  her payoff from a one-shot deviation followed by a return to TFT.

- **After**  $(C, C)$  (and at the empty history). Both play  $C$ : stream  $R, R, \dots$ , so  $\tilde{V} = R/(1 - \delta)$ . Deviating to  $D$  gives  $T$ , then the players are in the “ $(D, C)$ ” regime below, which yields the alternating stream:  $\tilde{V}^{\text{dev}} = T + \delta \frac{S + \delta T}{1 - \delta^2} = \frac{T + \delta S}{1 - \delta^2}$ .
- **After**  $(C, D)$  (player 1 was cheated). TFT makes player 1 play  $D$  and player 2 play  $C$ ; thereafter the roles alternate: stream  $T, S, T, S, \dots$ , so  $\tilde{V} = (T + \delta S)/(1 - \delta^2)$ . Deviating to  $C$  produces  $(C, C)$  and thus permanent mutual cooperation:  $\tilde{V}^{\text{dev}} = R/(1 - \delta)$ .
- **After**  $(D, C)$  (player 1 cheated). Player 1 plays  $C$ , player 2 plays  $D$ : stream  $S, T, S, T, \dots$ , so  $\tilde{V} = (S + \delta T)/(1 - \delta^2)$ . Deviating to  $D$  produces  $(D, D)$  and thus permanent mutual defection:  $\tilde{V}^{\text{dev}} = P/(1 - \delta)$ .
- **After**  $(D, D)$ . Both play  $D$  forever:  $\tilde{V} = P/(1 - \delta)$ . Deviating to  $C$  produces  $(C, D)$ , then  $(D, C)$ , then  $(C, D)$ , ...:  $\tilde{V}^{\text{dev}} = (S + \delta T)/(1 - \delta^2)$ .

**Theorem 4.9 (Knife-edge subgame perfection).** (TFT, TFT) is a subgame-perfect equilibrium of  $G^\infty(\delta)$  if and only if

$$\delta = \frac{T - R}{R - S} = \frac{P - S}{T - P}. \quad (4.6)$$

In particular, for generic payoffs satisfying (2.1)–(2.2) there is *no*  $\delta \in [0, 1]$  for which (TFT, TFT) is subgame perfect; for Axelrod's  $(5, 3, 1, 0)$  the two required values are  $2/3$  and  $1/4$ , so (TFT, TFT) is never subgame perfect.

*Proof.* By Theorem 4.3 we require  $\tilde{V}(h) \geq \tilde{V}^{\text{dev}}(h)$  in each of the four regimes above; by the symmetry of the profile these four conditions, applied to player 1, exhaust all histories and both players. Throughout,  $(1 - \delta) > 0$  and  $(1 - \delta^2) > 0$ , so multiplying through is legitimate.

$$\text{After } (C, C): \frac{R}{1 - \delta} \geq \frac{T + \delta S}{1 - \delta^2} \iff R(1 + \delta) \geq T + \delta S \iff \delta \geq \frac{T - R}{R - S}.$$

$$\text{After } (C, D): \frac{T + \delta S}{1 - \delta^2} \geq \frac{R}{1 - \delta} \iff T + \delta S \geq R(1 + \delta) \iff \delta \leq \frac{T - R}{R - S}.$$

$$\text{After } (D, C): \frac{S + \delta T}{1 - \delta^2} \geq \frac{P}{1 - \delta} \iff S + \delta T \geq P(1 + \delta) \iff \delta(T - P) \geq P - S \iff \delta \geq \frac{P - S}{T - P}.$$

$$\text{After } (D, D): \frac{P}{1 - \delta} \geq \frac{S + \delta T}{1 - \delta^2} \iff P(1 + \delta) \geq S + \delta T \iff \delta \leq \frac{P - S}{T - P}.$$

The first two conditions hold simultaneously iff  $\delta = (T - R)/(R - S)$ ; the last two iff  $\delta = (P - S)/(T - P)$ . Both pairs must hold, giving (4.6). Conversely, if (4.6) holds then all four inequalities hold with equality, no one-shot deviation is strictly profitable, and Theorem 4.3 gives subgame perfection. Writing  $x = T - R$ ,  $y = R - P$ ,  $z = P - S$  (all positive), (4.6) reads  $x/(y + z) = z/(x + y)$ , i.e.  $x(x + y) = z(z + y)$ , i.e.  $(x - z)(x + y + z) = 0$ , i.e.  $x = z$ : **(4.6) holds if and only if  $T - R = P - S$**  (equivalently  $T + S = R + P$ ), a single linear equation that fails on an open dense set of admissible payoffs. For  $(5, 3, 1, 0)$ :  $(T - R)/(R - S) = 2/3$  and  $(P - S)/(T - P) = 1/4$ , which are unequal.  $\square$

The two obstructions have distinct interpretations. The pair “after  $(C, C)$  / after  $(C, D)$ ” says that TFT’s retaliation must be *exactly* worth it: if  $\delta > (T - R)/(R - S)$ , punishing a defection by defecting locks the pair into the costly alternating cycle when immediate forgiveness would restore  $R$  forever, so the threat is not credible. The pair “after  $(D, C)$  / after  $(D, D)$ ” says the same about TFT’s behaviour in mutual defection: with  $\delta > (P - S)/(T - P)$  a player should offer an olive branch, absorbing  $S$  once to escape the  $P$ -forever trap, so unconditional continuation of the feud is not credible either.

The numbers for  $(5, 3, 1, 0)$  are collected in Table 4.1; the conditions are generated by `analytic.tft_spe_conditions_symbolic()` and evaluated by `analytic.thresholds()`.

Table 1: **Table 4.1.** One-shot-deviation checks for (TFT, TFT) under  $(T, R, P, S) = (5, 3, 1, 0)$ . Numerical values are unnormalised continuation payoffs at  $\delta = 2/3$ , the smallest  $\delta$  satisfying the on-path condition.

| Last profile | $\tilde{V}$ (follow TFT)            | $\tilde{V}^{\text{dev}}$ (one-shot deviation) | TFT optimal iff                             | at $\delta = 2/3$ |
|--------------|-------------------------------------|-----------------------------------------------|---------------------------------------------|-------------------|
| $(C, C)$     | $\frac{R}{1-\delta} = 9$            | $\frac{T+\delta S}{1-\delta^2} = 9$           | $\delta \geq \frac{T-R}{R-S} = \frac{2}{3}$ | holds (tie)       |
| $(C, D)$     | $\frac{T+\delta S}{1-\delta^2} = 9$ | $\frac{R}{1-\delta} = 9$                      | $\delta \leq \frac{T-R}{R-S} = \frac{2}{3}$ | holds (tie)       |
| $(D, C)$     | $\frac{S+\delta T}{1-\delta^2} = 6$ | $\frac{P}{1-\delta} = 3$                      | $\delta \geq \frac{P-S}{T-P} = \frac{1}{4}$ | holds             |
| $(D, D)$     | $\frac{P}{1-\delta} = 3$            | $\frac{S+\delta T}{1-\delta^2} = 6$           | $\delta \leq \frac{P-S}{T-P} = \frac{1}{4}$ | <b>fails</b>      |

**Remark 4.10 (Grim is perfect but brittle).** Theorems 4.5 and 4.9 pull in opposite directions. GRIM is subgame perfect for all  $\delta \geq 1/2$ , because its punishment is a stage equilibrium and hence self-enforcing; but the same absorbing punishment makes it catastrophic under noise: a single implementation error  $\varepsilon$  (Chapter 7) sends a GRIM pair into permanent mutual defection, so its long-run payoff per round is close to  $P$  for every  $\varepsilon > 0$  (the same as ALLD self-play,  $P + (T + S - 2P)\varepsilon - (T + S - R - P)\varepsilon^2 = 1 + 3\varepsilon - \varepsilon^2$ , i.e. 1.0299 at  $\varepsilon = 0.01$  under  $(5, 3, 1, 0)$ ; Chapter 7). TFT is forgiving but, as Theorem 4.9 shows, its threat is not credible — and under noise a TFT pair falls into an echo of alternating recriminations. As a memory-one vector TFT is  $(p_{CC}, p_{CD}, p_{DC}, p_{DD}) = (1, 0, 1, 0)$ , and the induced chain on  $(CC, CD, DC, DD)$  has, for *every*  $\varepsilon \in (0, \frac{1}{2}]$ , the uniform stationary distribution  $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ ; the long-run payoff is therefore  $(R + S + T + P)/4$ , which is 2.25 for  $(5, 3, 1, 0)$  rather than  $R = 3$ , no matter how small the error rate (Molander, 1985).

Two classical repairs, used later in the numerical chapters, address exactly this. *Contrite TIT FOR TAT* (Boerlijst et al., 1997) augments TFT with a standing label: a player who defects while the opponent is in good standing loses her own good standing, and a player in bad standing accepts one round of punishment without retaliating. A defection caused by error is thus atoned for in a single round instead of echoing, which makes the pair error-correcting while retaining TFT’s resistance to deliberate exploitation. *Win-Stay, Lose-Shift* (Pavlov) (Nowak & Sigmund, 1993) repeats its previous action after the two high payoffs  $T$  and  $R$  and switches after  $S$  and  $P$ ; as a memory-one vector it is  $(p_{CC}, p_{CD}, p_{DC}, p_{DD}) = (1, 0, 0, 1)$ , and a single error in a WSLS pair is corrected within two rounds. We state these without proof here; their stationary payoffs under noise are computed exactly in Chapter 7 via the four-state Markov chain of `pdlab.markov`.

## 4.5 The folk theorems

**Definition 4.11 (Feasible set).** The *feasible payoff set* of the repeated game is

$$V = \text{conv}\{(R, R), (S, T), (T, S), (P, P)\} \subset \mathbb{R}^2,$$

as in (2.3). Every normalised discounted payoff pair  $(U_1(s), U_2(s))$  lies in  $V$ , being a convex combination (with weights  $(1 - \delta)\delta^t$ ) of stage payoff vectors; conversely, with a public randomisation device every point of  $V$  is attainable, and without one every point of  $V$  is attainable in the limit  $\delta \rightarrow 1$ .

**Lemma 4.12 (Full dimensionality).**  $V$  has non-empty interior in  $\mathbb{R}^2$ .

*Proof.* The three points  $(R, R)$ ,  $(T, S)$ ,  $(S, T)$  are affinely independent: the determinant of the matrix with rows  $(T, S) - (R, R) = (T - R, S - R)$  and  $(S, T) - (R, R) = (S - R, T - R)$  equals  $(T - R)^2 - (S - R)^2 = (T - S)(T + S - 2R)$ , which is non-zero because  $T > S$  and, by (2.2),  $T + S - 2R < 0$ . Hence  $V$  contains a non-degenerate triangle.  $\square$

Thus the additivity condition (2.2), introduced in Remark 2.3 to make  $(C, C)$  efficient, is exactly what guarantees the full-dimensionality hypothesis of the subgame-perfect folk theorem.

**Definition 4.13 (Minmax and individual rationality).** Player  $i$ 's (*pure-action*) *minmax payoff* in the stage game is

$$\underline{v}_i = \min_{a_{-i} \in A_{-i}} \max_{a_i \in A_i} u_i(a_i, a_{-i}).$$

A payoff vector  $v$  is *individually rational* if  $v_i \geq \underline{v}_i$  for both  $i$ , and *strictly individually rational* if  $v_i > \underline{v}_i$  for both. Write  $V^* = \{v \in V : v_i > \underline{v}_i, i = 1, 2\}$ .

**Proposition 4.14 (Minmax of the Prisoner's Dilemma).**  $\underline{v}_1 = \underline{v}_2 = P$ , and the minmaxing action is  $D$ . The same value obtains if the minimum is taken over mixed strategies of the opponent.

*Proof.*  $\max_{a_1} u_1(a_1, C) = \max\{R, T\} = T$  and  $\max_{a_1} u_1(a_1, D) = \max\{S, P\} = P$ ; the minimum of  $\{T, P\}$  is  $P$  since  $T > P$ . For mixed  $\sigma_2$  with  $\sigma_2(C) = r$ ,  $\max_{a_1} u_1(a_1, \sigma_2) \geq u_1(D, \sigma_2) = rT + (1 - r)P \geq P$ , with equality at  $r = 0$ ; so the mixed minmax is also  $P$ .  $\square$

Consequently  $V^*$  is the set of feasible payoff pairs giving both players strictly more than mutual defection — the shaded region of Figure 1. It is non-empty (it contains  $(R, R)$ ) and, by Lemma 4.12, two-dimensional.

**Theorem 4.15 (Nash-threats folk theorem; Friedman 1971).** Let  $e = (P, P)$  be the payoff of the unique stage Nash equilibrium, and let  $v \in V$  satisfy  $v_i > P$  for  $i = 1, 2$ . Then there exists  $\bar{\delta} \in [0, 1)$  such that for every  $\delta \in (\bar{\delta}, 1)$  the supgame  $G^\infty(\delta)$  — extended by a public randomisation device if  $v$  is not a stage payoff vector — has a subgame-perfect equilibrium with normalised payoff vector  $v$  (Friedman, 1971).

*Proof sketch (the Friedman construction).* Let  $\alpha$  be a (possibly randomised) sequence or public lottery over action profiles whose normalised discounted payoff is  $v$ . Consider the trigger profile: play  $\alpha$  as long as no player has deviated from  $\alpha$ ; after any deviation, play  $D$  forever. Since  $(D, D)$  is a stage Nash equilibrium, the punishment phase is absorbing and prescribes a stage equilibrium at every history, so no one-shot deviation is profitable there (as in the punishment case of Theorem 4.5). Set  $M := \max_{i,a} u_i(a) - \min_{i,a} u_i(a) < \infty$ . In unnormalised terms a one-shot deviation in the cooperative phase gains at most  $M$  in the current round and forfeits, from the next round on, a continuation stream worth  $v_i$  per round in favour of one worth  $P$  per round, a loss of at least

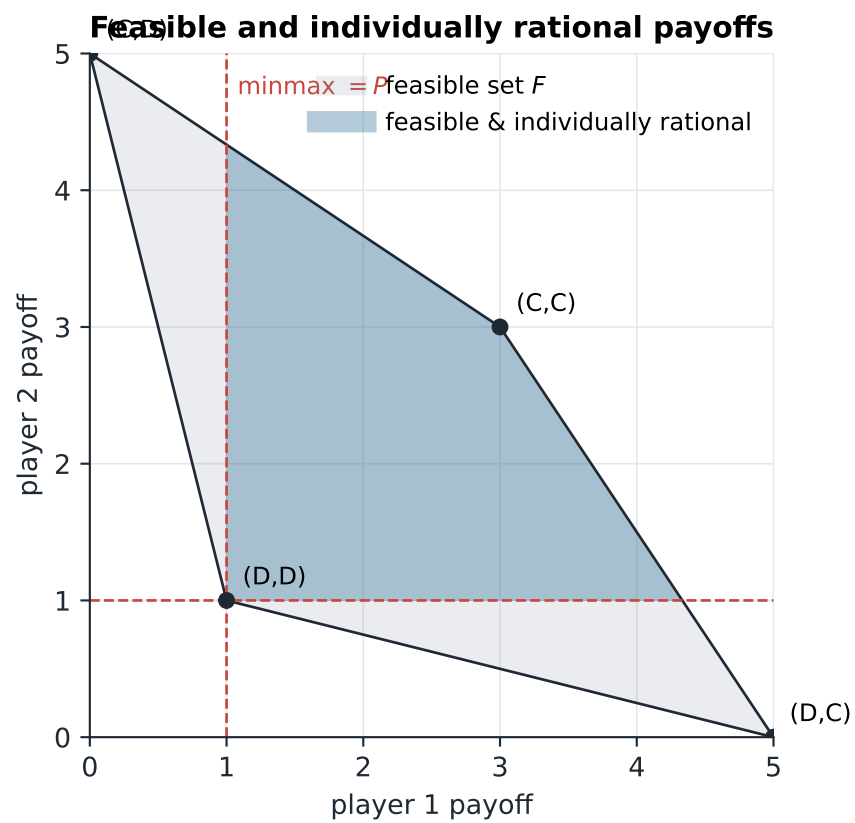


Figure 1: Feasible and individually rational payoff set for  $(5,3,1,0)$ .

$\frac{\delta}{1-\delta} \min_i (v_i - P) > 0$ . Since  $\min_i (v_i - P) > 0$  is fixed and  $\delta/(1-\delta) \rightarrow \infty$  as  $\delta \rightarrow 1$ , there is  $\bar{\delta} < 1$  beyond which the loss exceeds the gain at every history in the cooperative phase. Theorem 4.3 then yields subgame perfection.  $\square$

Friedman's theorem delivers everything strictly above the *stage-equilibrium* payoff. For the Prisoner's Dilemma this happens to coincide with everything strictly above the minmax, because  $\underline{v}_i = P$  is itself a stage equilibrium payoff — a coincidence special to this game. In general the two differ, and closing the gap requires punishments harsher than a stage equilibrium, together with rewards for punishers who would otherwise refuse to punish.

**Theorem 4.16 (Subgame-perfect folk theorem; Fudenberg and Maskin 1986).** Let  $G$  be a finite  $n$ -player stage game whose feasible set  $V \subset \mathbb{R}^n$  has full dimension  $n$ , and let  $\underline{v}_i^{\text{mix}}$  denote player  $i$ 's *mixed* minmax payoff (the opponents may correlate on mixed actions; it is weakly below the pure minmax of Definition 4.13, and equal to it in the Prisoner's Dilemma by Proposition 4.14). Then for every  $v \in V$  with  $v_i > \underline{v}_i^{\text{mix}}$  for all  $i$  there exists  $\bar{\delta} \in [0, 1)$  such that for all  $\delta \in (\bar{\delta}, 1)$  the discounted supgame has a subgame-perfect equilibrium with normalised payoff vector  $v$  (Fudenberg & Maskin, 1986).

Full dimensionality is used to construct, for each player  $i$ , a continuation payoff vector that rewards the other players  $j \neq i$  for carrying out  $i$ 's punishment while still leaving  $i$  punished — impossible if the feasible set is degenerate, for instance in a zero-sum game. Fudenberg and Maskin also show that for  $n = 2$  the dimensionality hypothesis can be dispensed with. For the Prisoner's Dilemma, Lemma 4.12 shows the hypothesis holds anyway, and Proposition 4.14 identifies  $V^*$  concretely: for  $(5, 3, 1, 0)$ , every payoff pair in the quadrilateral with vertices  $(3, 3), (0, 5), (5, 0), (1, 1)$  that gives both players strictly more than 1 is a subgame-perfect equilibrium payoff for  $\delta$  close enough to 1.

**Remark 4.17 (What the folk theorems do and do not say).** They say that patience makes cooperation *possible*, not that it makes cooperation *predicted*: for  $\delta$  near 1 the equilibrium set is enormous, containing mutual defection, mutual cooperation, and every asymmetric split in between. Equilibrium selection is therefore left entirely open, which is the intellectual justification for the rest of this report. Axelrod's tournaments (Chapter 6) select among strategies by ecological success rather than by refinement; evolutionary game theory (Chapter 5) selects by dynamic stability; and reinforcement learning (Chapter 7) selects by what adaptive agents actually converge to. None of these is a refinement of subgame perfection, and they need not agree — which is precisely what makes the comparison worth making.

## 5 Evolutionary Game Theory

Chapters 3 and 4 asked which strategy profiles a rational player would choose. This chapter asks a different question: which strategies *survive*. Payoffs become reproductive fitness, deliberation is replaced by differential reproduction or imitation, and equilibrium is replaced by dynamic stability.

### 5.1 The replicator equation and evolutionary stability

**Definition 5.1 (Population game).** Let  $\mathcal{S} = \{1, \dots, n\}$  be a finite set of pure strategies and  $A = (a_{ij}) \in \mathbb{R}^{n \times n}$  a payoff matrix,  $a_{ij}$  being the payoff to an  $i$ -player matched with a  $j$ -player. A *population state* is a point of the simplex

$$\Delta^{n-1} = \left\{ x \in \mathbb{R}_{\geq 0}^n : \sum_i x_i = 1 \right\},$$

$x_i$  being the frequency of strategy  $i$ . In a randomly matched infinite population, the expected payoff to strategy  $i$  is  $f_i(x) = (Ax)_i$  and the mean population payoff is  $\bar{f}(x) = x^\top Ax$ .

**Definition 5.2 (Replicator equation; Taylor and Jonker 1978).** The replicator dynamics on  $\Delta^{n-1}$  is

$$\dot{x}_i = x_i[(Ax)_i - x^\top Ax], \quad i = 1, \dots, n. \quad (5.1)$$

(Taylor & Jonker, 1978) A strategy grows exactly when it earns more than the population average.

The simplex and each of its faces are invariant:  $\sum_i \dot{x}_i = \bar{f} - \bar{f} = 0$ , and  $x_i = 0 \Rightarrow \dot{x}_i = 0$ . Adding a constant to any *column* of  $A$  leaves (5.1) unchanged, a fact used below to simplify computations. Equation (5.1) is implemented as `analytic.replicator_rhs`.

**Definition 5.3 (Evolutionarily stable strategy; Maynard Smith and Price 1973).** A state  $x^* \in \Delta^{n-1}$  is an *evolutionarily stable strategy* (ESS) if for every  $y \neq x^*$  there is  $\bar{\epsilon} \in (0, 1)$  such that

$$(x^*)^\top A(\epsilon y + (1 - \epsilon)x^*) > y^\top A(\epsilon y + (1 - \epsilon)x^*) \quad \text{for all } \epsilon \in (0, \bar{\epsilon}).$$

(Maynard Smith & Price, 1973) Equivalently (expanding in  $\epsilon$ ),  $x^*$  is an ESS iff for every  $y \neq x^*$ :

$$\text{(E1)} \quad (x^*)^\top Ax^* \geq y^\top Ax^*, \quad \text{and} \quad \text{(E2)} \quad \text{if } (x^*)^\top Ax^* = y^\top Ax^* \text{ then } (x^*)^\top Ay > y^\top Ay. \quad (5.2)$$

If the strict inequality in (E2) is weakened to “ $\geq$ ”,  $x^*$  is called *neutrally stable* (an NSS).

For a pure strategy  $i$ , (5.2) reads: for every  $j \neq i$ , either  $a_{ii} > a_{ji}$ , or  $a_{ii} = a_{ji}$  and  $a_{ij} > a_{jj}$ . These are the conditions tested by `analytic.is_ess` and, in the weakened form, `analytic.is_neutrally_stable`.

**Theorem 5.4 (ESS, Nash, and stability).** Let  $x^*$  be an ESS of the population game  $A$ . Then (i)  $(x^*, x^*)$  is a symmetric Nash equilibrium of the underlying two-player symmetric game, and (ii)  $x^*$  is an asymptotically stable rest point of the replicator dynamics (5.1). Neither converse holds in general: a symmetric Nash equilibrium need not be an ESS (e.g. TFT in Theorem 5.10), and for  $n \geq 3$  strategies there are asymptotically stable rest points that are not evolutionarily stable; for  $n = 2$  the two notions coincide.

*Proof.* (i) is condition (E1), which is precisely the statement that no  $y$  is a strictly better reply to  $x^*$  than  $x^*$  itself. For (ii) see Hofbauer and Sigmund (Hofbauer & Sigmund, 1998, Thm. 7.2.4): the relative entropy  $H(x) = -\sum_{i \in \text{supp } x^*} x_i^* \log(x_i/x_i^*)$  is a strict local Lyapunov function for (5.1) at an ESS, since  $\dot{H} = -[(x^*)^\top Ax - x^\top Ax] < 0$  in a punctured neighbourhood of  $x^*$  by (5.2).  $\square$

We shall also need the eigenvalues of (5.1) at the vertices.

**Lemma 5.5 (Vertex spectrum).** In the coordinates  $(x_j)_{j \neq i}$  on  $\Delta^{n-1}$  (with  $x_i = 1 - \sum_{j \neq i} x_j$ ), the Jacobian of (5.1) at the vertex  $e_i$  is diagonal with entries

$$\lambda_j = a_{ji} - a_{ii}, \quad j \neq i.$$

*Proof.* For  $j \neq i$  write  $\dot{x}_j = x_j G_j(x)$  with  $G_j(x) = (Ax)_j - x^\top Ax$ . Then  $\partial \dot{x}_j / \partial x_k = \delta_{jk} G_j(x) + x_j \partial G_j / \partial x_k$ . At  $x = e_i$  we have  $x_j = 0$  for all  $j \neq i$ , killing the second term, and  $G_j(e_i) = a_{ji} - a_{ii}$ .  $\square$

Thus  $e_i$  is asymptotically stable iff  $a_{ji} < a_{ii}$  for all  $j \neq i$  (a *strict* symmetric Nash equilibrium), which for pure strategies is also exactly the ESS condition without the tie-breaking clause.

## 5.2 The one-shot dilemma: ALLD is the unique ESS

Take  $\mathcal{S} = \{C, D\}$  with  $A = \begin{pmatrix} R & S \\ T & P \end{pmatrix}$  and write  $x$  for the frequency of  $C$ .

**Theorem 5.6.** In the one-shot Prisoner's Dilemma the replicator dynamics on  $\Delta^1$  is

$$\dot{x} = x(1-x)[x(R-T) + (1-x)(S-P)],$$

which is strictly negative for all  $x \in (0, 1)$ . Hence  $x(t) \rightarrow 0$  for every initial  $x(0) < 1$ , the vertex  $D$  is globally asymptotically stable on  $[0, 1)$ , and  $D$  is the unique ESS;  $C$  is not even neutrally stable.

*Proof.* Substituting into (5.1) with  $n = 2$  gives  $\dot{x} = x[(Ax)_C - \bar{f}] = x(1-x)[(Ax)_C - (Ax)_D]$ , and  $(Ax)_C - (Ax)_D = x(R-T) + (1-x)(S-P)$ . Both  $R-T$  and  $S-P$  are strictly negative by (2.1), so the bracket is a convex combination of two negative numbers, hence negative on  $[0, 1]$ ;  $\dot{x} < 0$  on  $(0, 1)$  gives monotone convergence to 0. By Lemma 5.5 the eigenvalue at  $e_D$  is  $a_{CD} - a_{DD} = S - P < 0$  and at  $e_C$  it is  $a_{DC} - a_{CC} = T - R > 0$ . ESS:  $a_{DD} = P > S = a_{CD}$  verifies (E1) strictly for  $D$ , whereas for  $C$  we have  $a_{CC} = R < T = a_{DC}$ , violating (E1) even weakly.  $\square$

This is the evolutionary restatement of Theorem 2.7: selection on individual fitness in a well-mixed population destroys cooperation. Everything that follows is an account of what must be added to the model to prevent this.

## 5.3 Repeated play: the system $\{\text{ALLC}, \text{ALLD}, \text{TFT}\}$

Suppose now that each encounter is not a single round but a discounted supgame  $G^\infty(\delta)$ , and that the heritable unit is a repeated-game strategy. Restrict attention to the three strategies  $\mathcal{S}_3 = (\text{ALLC}, \text{ALLD}, \text{TFT})$ , indexed 1, 2, 3 in that order, with payoffs given by the *normalised* discounted value (4.1) so that entries are directly comparable to stage payoffs.

**Proposition 5.7 (The  $3 \times 3$  payoff matrix).** With normalised discounted payoffs,

$$A(\delta) = \begin{pmatrix} R & S & R \\ T & P & (1-\delta)T + \delta P \\ R & (1-\delta)S + \delta P & R \end{pmatrix}. \quad (5.3)$$

*Proof.* Each entry is  $(1-\delta) \sum_{t \geq 0} \delta^t \pi_t$  for the realised path.

*Row ALLC.* Against ALLC the path is  $(C, C)$  forever:  $R$ . Against ALLD it is  $(C, D)$  forever:  $S$ . Against TFT: TFT opens with  $C$  and thereafter copies  $C$ , so the path is  $(C, C)$  forever:  $R$ .

*Row ALLD.* Against ALLC:  $T$  forever. Against ALLD:  $P$  forever. Against TFT: round 0 is  $(D, C)$  giving  $T$ ; from round 1 onward TFT plays  $D$  and the path is  $(D, D)$ . Hence  $(1-\delta)T + (1-\delta) \sum_{t \geq 1} \delta^t P = (1-\delta)T + \delta P$ .

*Row TFT.* Against ALLC:  $(C, C)$  forever,  $R$ . Against ALLD: round 0 is  $(C, D)$  giving  $S$ , then  $(D, D)$  forever; hence  $(1-\delta)S + \delta P$ . Against TFT: mutual cooperation,  $R$ .  $\square$

This is `analytic.payoff_matrix_3_symbolic()`. For  $(5, 3, 1, 0)$  and  $\delta = 0.9$  it evaluates to

$$A(0.9) = \begin{pmatrix} 3 & 0 & 3 \\ 5 & 1 & 1.4 \\ 3 & 0.9 & 3 \end{pmatrix}.$$

Two structural features of (5.3) drive everything below. First, ALLC and TFT are *payoff-indistinguishable* against each other and against themselves: rows 1 and 3 agree in columns 1 and 3, and columns 1 and 3 agree in every row. In a population containing no ALLD, the two strategies are behaviourally identical — TFT never gets to retaliate — and selection cannot separate them. Second, TFT differs from ALLC only in column 2:  $(1 - \delta)S + \delta P > S$  for  $\delta > 0$ . TFT's entire evolutionary advantage is that it loses to ALLD only once.

### 5.3.1 Rest points

**Theorem 5.8 (Rest points).** Assume (2.1)–(2.2) and  $\delta \in (0, 1)$ . Set

$$\sigma := (1 - \delta)(P - S) > 0, \quad \Delta := (R - P) - (1 - \delta)(T - P). \quad (5.4)$$

The rest points of (5.1) with  $A = A(\delta)$  are: the three vertices; every point of the ALLC–TFT edge  $\{(x_1, 0, x_3) : x_1 + x_3 = 1\}$ ; and, when  $\Delta > 0$ , the single point

$$x^\dagger = (0, 1 - x^*, x^*), \quad x^* = \frac{\sigma}{\sigma + \Delta} = \frac{(1 - \delta)(P - S)}{(1 - \delta)(P - S) + (R - P) - (1 - \delta)(T - P)} \quad (5.5)$$

on the ALLD–TFT edge. Moreover  $x^* \in (0, 1)$  if and only if  $\Delta > 0$ , i.e. if and only if

$$\delta > \frac{T - R}{T - P} = \delta^G,$$

the Grim/Nash threshold (4.2). There are no rest points in the interior of the simplex.

*Proof. Edge ALLC–TFT.* On  $x_2 = 0$ ,  $(Ax)_1 = Rx_1 + Rx_3 = R$  and  $(Ax)_3 = R$ , so  $\bar{f} = R$  and  $\dot{x}_1 = \dot{x}_3 = 0$ : the entire edge consists of rest points. This is the payoff-indistinguishability noted above.

*Edge ALLD–TFT.* Put  $x = (0, 1 - x_3, x_3)$ . Then

$$(Ax)_2 = x_3[(1 - \delta)T + \delta P] + (1 - x_3)P, \quad (Ax)_3 = x_3R + (1 - x_3)[(1 - \delta)S + \delta P],$$

and a rest point with  $x_3 \in (0, 1)$  requires  $(Ax)_2 = (Ax)_3$ . Rearranging,

$$x_3 \left[ R - (1 - \delta)T - \delta P \right] = (1 - x_3) \left[ P - (1 - \delta)S - \delta P \right],$$

i.e.  $x_3 \Delta = (1 - x_3) \sigma$ , since  $R - (1 - \delta)T - \delta P = (R - P) - (1 - \delta)(T - P) = \Delta$  and  $P - (1 - \delta)S - \delta P = (1 - \delta)(P - S) = \sigma$ . Solving gives  $x_3 = \sigma / (\sigma + \Delta) = x^*$ . As  $\sigma > 0$ ,  $x^* \in (0, 1)$  iff  $\Delta > 0$ ; and  $\Delta > 0 \iff (R - P) > (1 - \delta)(T - P) \iff \delta > (T - R) / (T - P)$  (divide by  $T - P > 0$ ).

*Edge ALLC–ALLD.* Here  $(Ax)_1 = Rx_1 + Sx_2$  and  $(Ax)_2 = Tx_1 + Px_2$ , and  $(Ax)_2 - (Ax)_1 = x_1(T - R) + x_2(P - S) > 0$  whenever  $x \neq e_{\text{ALLC}}$ ; there is no interior rest point on this edge (this is Theorem 5.6 again).

*Interior.* An interior rest point requires  $(Ax)_1 = (Ax)_2 = (Ax)_3$ . But  $(Ax)_3 - (Ax)_1 = x_2[(1 - \delta)S + \delta P - S] = x_2\delta(P - S) > 0$  for  $x_2 > 0$ , so TFT strictly outperforms ALLC at every state with  $x_2 > 0$ : no interior rest point exists.  $\square$

For  $(5, 3, 1, 0)$  and  $\delta = 0.9$ :  $\sigma = 0.1$ ,  $\Delta = 2 - 0.1 \cdot 4 = 1.6$ , hence  $x^* = 0.1/1.7 = 1/17 \approx 0.0588$ , reproduced by `analytic.tft_invasion_threshold_symbolic()`.

### 5.3.2 Stability

**Theorem 5.9 (Spectra and classification).** Let  $\delta \in (0, 1)$ ,  $\sigma, \Delta$  as in (5.4), and use the simplex coordinates  $(x_1, x_3) = (x_{\text{ALLC}}, x_{\text{TFT}})$ . Then the Jacobian eigenvalues of (5.1) are:

| Rest point                       | eigenvalues                                            | classification                           |
|----------------------------------|--------------------------------------------------------|------------------------------------------|
| ALLC                             | $T - R (> 0), \quad 0$                                 | unstable                                 |
| ALLD                             | $-(P - S) (< 0), \quad -\sigma (< 0)$                  | asymptotically stable                    |
| TFT                              | $-\Delta, \quad 0$                                     | stable (non-hyperbolic) iff $\Delta > 0$ |
| $x^\dagger$ (when $\Delta > 0$ ) | $-\delta(P - S)(1 - x^*) (< 0), \quad x^*\Delta (> 0)$ | saddle                                   |

*Proof.* The three vertex rows follow from Lemma 5.5 applied to (5.3):

- At  $e_{\text{ALLC}}$ :  $a_{21} - a_{11} = T - R$  and  $a_{31} - a_{11} = R - R = 0$ .
- At  $e_{\text{ALLD}}$ :  $a_{12} - a_{22} = S - P = -(P - S)$  and  $a_{32} - a_{22} = (1 - \delta)S + \delta P - P = -(1 - \delta)(P - S) = -\sigma$ .
- At  $e_{\text{TFT}}$ :  $a_{13} - a_{33} = R - R = 0$  and  $a_{23} - a_{33} = (1 - \delta)T + \delta P - R = -\Delta$ .

For  $x^\dagger$ , write  $g(x_1, x_3) = (\dot{x}_1, \dot{x}_3)$  with  $x_2 = 1 - x_1 - x_3$  and differentiate; the computation is carried out symbolically by `analytic.jacobian_eigenvalues_symbolic("TFT_ALLD_edge")`, which returns

$$\lambda_1 = \frac{\delta(P - S)[(T - R) - \delta(T - P)]}{\sigma + \Delta}, \quad \lambda_2 = \frac{-(1 - \delta)(P - S)[(T - R) - \delta(T - P)]}{\sigma + \Delta}.$$

Since  $(T - R) - \delta(T - P) = -[(R - P) - (1 - \delta)(T - P)] = -\Delta$ ,  $(1 - \delta)(P - S) = \sigma$ , and  $\sigma/(\sigma + \Delta) = x^*$ , these simplify to

$$\lambda_1 = -\delta(P - S) \frac{\Delta}{\sigma + \Delta} = -\delta(P - S)(1 - x^*), \quad \lambda_2 = \frac{\sigma \Delta}{\sigma + \Delta} = x^* \Delta.$$

When  $\Delta > 0$  we have  $\lambda_1 < 0 < \lambda_2$ , a hyperbolic saddle. The eigenvector for  $\lambda_2$  lies along the ALLD–TFT edge and the one for  $\lambda_1$  points into the simplex; the latter sign is intuitive, since at  $x^\dagger$  strategy ALLC earns  $x^*R + (1 - x^*)S$ , which is strictly less than TFT's  $x^*R + (1 - x^*)[(1 - \delta)S + \delta P]$ , so the ALLC direction is contracting.

For the classification of  $e_{\text{TFT}}$ , note that the zero eigenvalue is not an artefact: the whole ALLC–TFT edge consists of rest points (Theorem 5.8), so  $e_{\text{TFT}}$  can be Lyapunov stable but never asymptotically stable. It *is* Lyapunov stable when  $\Delta > 0$ , since the transverse eigenvalue  $-\Delta$  is negative and the centre direction is a line of rest points.  $\square$

For  $(5, 3, 1, 0)$ ,  $\delta = 0.9$  the four spectra are  $\{2, 0\}$ ,  $\{-1, -0.1\}$ ,  $\{-1.6, 0\}$  and  $\{-0.847, +0.0941\}$ , matching the numerical Jacobian of `analytic.jacobian_eigenvalues` to eight digits.

**Theorem 5.10 (ESS classification).** Let  $\delta \in (0, 1)$ .

1. ALLD is an ESS, for every  $\delta \in (0, 1)$ .
2. TFT is *not* an ESS, for any  $\delta$ ; it is neutrally stable if and only if  $\delta > (T - R)/(T - P)$ .
3. ALLC is neither an ESS nor neutrally stable.

*Proof.* (1) Compare column ALLD:  $a_{22} = P$ ,  $a_{12} = S < P$ , and  $a_{32} = (1 - \delta)S + \delta P = P - \sigma < P$ . Both inequalities are strict, so (E1) holds strictly for both  $y = e_{\text{ALLC}}$  and  $y = e_{\text{TFT}}$ , and by linearity for every mixture  $y \neq e_{\text{ALLD}}$ . Hence ALLD is a strict Nash equilibrium and therefore an ESS.

(2) Take  $y = e_{\text{ALLC}}$ . Then  $a_{33} = R = a_{13}$ , so (E1) holds with equality and (E2) is required: it demands  $a_{31} > a_{11}$ , i.e.  $R > R$ , which is false. So TFT fails (E2) and is not an ESS. For neutral stability (E2) is weakened to  $a_{31} \geq a_{11}$ , i.e.  $R \geq R$ : true. The remaining comparison is  $y = e_{\text{ALLD}}$ , requiring  $a_{33} \geq a_{23}$ , i.e.  $R \geq (1 - \delta)T + \delta P$ , i.e.  $\Delta \geq 0$ , i.e.  $\delta \geq (T - R)/(T - P)$ ; strict neutral stability against ALLD invasions needs the strict inequality because at  $\Delta = 0$  (E1) holds with equality and the second clause then demands  $a_{32} \geq a_{22}$ , i.e.  $(1 - \delta)S + \delta P \geq P$ , i.e.  $-\sigma \geq 0$ , which fails; and mixtures  $y$  with  $y_2 > 0$  then also fail to invade by linearity.

(3)  $a_{11} = R < T = a_{21}$ , so (E1) fails already for  $y = e_{\text{ALLD}}$ .  $\square$

**Remark 5.11 (Bistability and the invasion barrier).** Combining Theorems 5.9 and 5.10: for  $\delta > \delta^G = (T - R)/(T - P)$  the system is *bistable*. ALLD and the TFT-end of the cooperative edge are both stable, and the saddle  $x^\dagger$  is the barrier between their basins along the ALLD–TFT edge: a TFT mutant arising singly in an ALLD population is selected against, because a lone TFT pays  $S$  once against almost every opponent and never collects  $R$ ; only if the TFT frequency exceeds  $x^*$  do TFT–TFT encounters become common enough for reciprocity to pay. Reciprocity therefore requires a *critical mass*, supplied in practice by clustering, kin structure or drift — themes taken up in Sections 5.4 and 5.5. Note how small the barrier is for realistic patience:  $x^* = 1/17 \approx 5.9\%$  at  $\delta = 0.9$ , and  $x^* \rightarrow 0$  as  $\delta \rightarrow 1$  since  $\sigma \rightarrow 0$  while  $\Delta \rightarrow R - P > 0$ .

The second consequence is the fragility of the cooperative state. Because the ALLC–TFT edge is a line of rest points, a TFT population is subject to *neutral drift* towards ALLC: ALLC mutants pay no selective price while ALLD is absent. Once drift has raised  $x_{\text{ALLC}}$  far enough, a subsequent ALLD invasion succeeds, and the cycle repeats. This is the standard account of the observed oscillations between defection, reciprocity and unconditional cooperation in evolutionary simulations, and the reason more forgiving reciprocators such as Generous TIT FOR TAT (Nowak & Sigmund, 1992) and Win-Stay-Lose-Shift (Nowak & Sigmund, 1993) eventually displace TFT in longer-run models.

## 5.4 Finite populations: the Moran process

The replicator equation describes an infinite population; in a finite one, selection competes with sampling noise, and the relevant quantity is a *fixation probability*.

**Definition 5.12 (Moran process with frequency-dependent fitness).** (Nowak, 2006a, Chapter 6) Fix a population of constant size  $N$  containing  $i$  individuals of type  $A$  and  $N - i$  of type  $B$ . With payoff matrix  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$  (rows  $A, B$ ) and *no self-interaction*, the expected payoffs are

$$\pi_A(i) = \frac{a(i-1) + b(N-i)}{N-1}, \quad \pi_B(i) = \frac{c i + d(N-i-1)}{N-1},$$

and fitnesses are  $f_i = 1 - w + w \pi_A(i)$  and  $g_i = 1 - w + w \pi_B(i)$ , where  $w \in [0, 1]$  is the *intensity of selection* ( $w \rightarrow 0$  is *weak selection*,  $w = 1$  *strong*). In each elementary step one individual is chosen to reproduce with probability proportional to its fitness, its offspring replaces a uniformly chosen individual, and  $i$  changes by at most one. The resulting birth–death chain on  $\{0, 1, \dots, N\}$  has absorbing states 0 and  $N$  and transition probabilities

$$p_{i,i+1} = \frac{i f_i}{i f_i + (N-i) g_i} \cdot \frac{N-i}{N}, \quad p_{i,i-1} = \frac{(N-i) g_i}{i f_i + (N-i) g_i} \cdot \frac{i}{N}.$$

**Theorem 5.13 (Fixation probability).** The probability that a single  $A$  mutant introduced into a resident  $B$  population reaches fixation is

$$\rho_A = \frac{1}{1 + \sum_{k=1}^{N-1} \prod_{j=1}^k \frac{g_j}{f_j}}. \quad (5.6)$$

(Nowak, 2006a, eq. (6.20)) Under neutrality ( $w = 0$ , so  $f_j = g_j = 1$ ) this reduces to the *neutral benchmark*  $\rho_A = 1/N$ . A mutant is said to be *favoured by selection* if  $\rho_A > 1/N$ .

*Proof.* For a birth–death chain with absorbing boundaries the ratio  $\gamma_i := p_{i,i-1}/p_{i,i+1}$  equals  $g_i/f_i$  (the  $N$ -dependent factors cancel), and the classical gambler’s-ruin formula gives  $\rho_A = (1 + \sum_{k=1}^{N-1} \prod_{j=1}^k \gamma_j)^{-1}$ .  $\square$

**Theorem 5.14 (The 1/3 law; Nowak, Sasaki, Taylor and Fudenberg 2004).** Let the two-strategy game be a *coordination game*, i.e.  $a > c$  and  $d > b$ , so that the replicator dynamics has an unstable interior equilibrium at

$$x^* = \frac{d - b}{(a - c) + (d - b)}$$

(the frequency of  $A$  at which the two types earn equally). Then, for weak selection and large  $N$ ,

$$\rho_A > \frac{1}{N} \iff x^* < \frac{1}{3}.$$

(Nowak et al., 2004)

The result is initially surprising — the naive guess would be  $x^* < 1/2$  — and its source is that a mutant lineage on its way to fixation spends its time at low frequencies, so it experiences the payoff environment weighted towards the resident. Applied to the ALLD–TFT edge of Section 5.3, with  $A = \text{TFT}$  and  $B = \text{ALLD}$ , the barrier is  $x^* = \sigma/(\sigma + \Delta)$  from (5.5); at  $\delta = 0.9$  and  $(5, 3, 1, 0)$  this is  $1/17 \ll 1/3$ , so a single TFT mutant in an ALLD population is favoured by selection in a finite population even though the replicator dynamics of Theorem 5.9 sends it to extinction. Finite-population stochasticity is thus a *mechanism for cooperation* in its own right — the discrepancy between (5.6) and (5.1) is not an approximation error but a genuinely different prediction.

## 5.5 Spatial structure: the Nowak–May lattice

Cooperation can also be rescued by giving the population geometry, so that cooperators interact disproportionately with cooperators. The canonical demonstration is due to Nowak and May (Nowak & May, 1992, 1993).

**Definition 5.15 (Nowak–May spatial game).** Sites of an  $L \times L$  square lattice (periodic or fixed boundary) are each occupied by one player of type  $C$  or  $D$ . Payoffs are the *weak dilemma* of Definition 2.12:  $R = 1$ ,  $T = b$ ,  $P = S = 0$ , with  $1 < b < 2$ . In each generation:

1. **Scoring.** Every player plays one round against each of its eight nearest neighbours (Moore neighbourhood) *and against itself*, and accumulates the nine payoffs.
2. **Updating.** Synchronously, every site adopts the strategy of the highest-scoring player in its own  $3 \times 3$  neighbourhood (itself included); ties are resolved in favour of the incumbent.

Because  $P = S = 0$ , the scores take a particularly simple form. Write  $\nu(z)$  for the number of cooperators among the nine cells of  $z$ 's  $3 \times 3$  neighbourhood. Then

$$\text{score}(z) = \nu(z) \quad \text{if } z \text{ is } C, \quad \text{score}(z) = b\nu(z) \quad \text{if } z \text{ is } D. \quad (5.7)$$

(A cooperator's own self-interaction contributes  $R = 1$  and is counted in  $\nu$ ; a defector's contributes  $P = 0$  and is not.) All thresholds below follow from (5.7) by counting.

**Proposition 5.16 (A lone defector).** Let a single  $D$  sit in an otherwise all- $C$  lattice.

1. The defector always survives: its score is  $8b$ , the largest score in its own neighbourhood, since its eight neighbours are cooperators with  $\nu = 8$  and  $8b > 8$ .
2. Its eight neighbours convert to  $D$  if and only if  $b > 9/8$ . Hence for  $1 < b < 9/8$  the lone defector is frozen forever; for  $b > 9/8$  it grows into a  $3 \times 3$  block of defectors in a single generation.

*Proof.* Put the defector at the origin. A cooperator  $z$  adjacent to it (orthogonally or diagonally) has  $\nu(z) = 8$ , hence score 8. Its own  $3 \times 3$  neighbourhood contains the defector, with score  $8b$ ; it also contains at least one cooperator at Chebyshev distance 2 from the origin, whose neighbourhood excludes the origin and which therefore has score 9. (For an orthogonal neighbour at  $(0, 1)$  take the cell  $(-1, 2)$ ; for a diagonal neighbour at  $(1, 1)$  take  $(2, 2)$ .) No cooperator can score more than 9. So the maximum in  $z$ 's neighbourhood is  $\max\{8b, 9\}$ , and  $z$  becomes  $D$  exactly when  $8b > 9$ .  $\square$

**Proposition 5.17 (A  $3 \times 3$  block of defectors).** Let a  $3 \times 3$  block of  $D$  occupy rows and columns 0, 1, 2 of an otherwise all- $C$  lattice. Using (5.7), the relevant scores are:

| site                                                | $\nu$ | score |
|-----------------------------------------------------|-------|-------|
| block corner, e.g. $(0, 0)$ ( $D$ )                 | 5     | $5b$  |
| block edge-centre, e.g. $(0, 1)$ ( $D$ )            | 3     | $3b$  |
| block centre $(1, 1)$ ( $D$ )                       | 0     | 0     |
| $C$ diagonally off a block corner, $(-1, -1)$       | 8     | 8     |
| $C$ orthogonally off a block corner, $(-1, 0)$      | 7     | 7     |
| $C$ orthogonally off a block edge-centre, $(-1, 1)$ | 6     | 6     |
| $C$ at Chebyshev distance $\geq 2$ from the block   | 9     | 9     |

Consequently:

1. **The block expands if and only if  $b > 9/5 = 1.8$ .** Every cooperator adjacent to the block has a score-9 cooperator in its own neighbourhood, and the largest defector score available to it is  $5b$  (contributed by a block corner). Hence such a cooperator flips iff  $5b > 9$ .
2. **The block's corners survive if and only if  $b \geq 8/5 = 1.6$ .** The corner  $(0, 0)$  sees a maximum cooperator score of 8 (from the diagonal cooperator at  $(-1, -1)$ ) and a maximum defector score of  $5b$  (itself); it stays  $D$  iff  $5b \geq 8$  (ties favour the incumbent defector).
3. **The block's edges survive if and only if  $b \geq 7/5 = 1.4$ .** The edge-centre  $(0, 1)$  sees a maximum cooperator score of 7 (from  $(-1, 0)$  and  $(-1, 2)$ ) and a maximum defector score of  $5b$ ; it stays  $D$  iff  $5b \geq 7$ .

*Proof.* Each claim is the comparison of the largest  $C$ -score with the largest  $D$ -score in the relevant  $3 \times 3$  neighbourhood, read off the table.

For (1): the cooperator at  $(-1, 1)$  has neighbourhood rows  $-2..0$ , columns  $0..2$ , containing the defectors  $(0, 0)$ ,  $(0, 1)$ ,  $(0, 2)$  with scores  $5b, 3b, 5b$  and the cooperators  $(-2, 0)$ ,  $(-2, 1)$ ,  $(-2, 2)$  with score 9. The cooperator at  $(-1, -1)$  has neighbourhood rows  $-2..0$ , columns  $-2..0$ , containing the single defector  $(0, 0)$  with score  $5b$  and cooperators of score up to 9. In both cases the condition is  $5b > 9$ ; by symmetry the same holds around the whole block.

For (2):  $(0, 0)$ 's neighbourhood is rows  $-1..1$ , columns  $-1..1$ , containing cooperators  $(-1, -1)$ ,  $(-1, 0)$ ,  $(-1, 1)$ ,  $(0, -1)$ ,  $(1, -1)$  of scores 8, 7, 6, 7, 6 and defectors  $(0, 0)$ ,  $(0, 1)$ ,  $(1, 0)$ ,  $(1, 1)$  of scores  $5b, 3b, 3b, 0$ .

For (3):  $(0, 1)$ 's neighbourhood is rows  $-1..1$ , columns  $0..2$ , containing cooperators  $(-1, 0)$ ,  $(-1, 1)$ ,  $(-1, 2)$  of scores 7, 6, 7 and defectors  $(0, 0)$ ,  $(0, 1)$ ,  $(0, 2)$ ,  $(1, 0)$ ,  $(1, 1)$ ,  $(1, 2)$  of scores  $5b, 3b, 5b, 3b, 0, 3b$ .  $\square$

**Proposition 5.18 (A straight interface, and the role of  $b = 2$ ).** Let the half-plane of rows  $\leq -1$  be all  $D$  and rows  $\geq 0$  all  $C$ . Then row-0 cooperators score 6, cooperators in rows  $\geq 1$  score 9, row-(-1) defectors score  $3b$ , and defectors in rows  $\leq -2$  score 0. Consequently:

1. The cooperative front never retreats: a row-0 cooperator sees a maximum cooperator score of 9 and a maximum defector score of  $3b < 6 < 9$ .
2. The cooperative front advances — the row-(-1) defectors convert — if and only if  $6 > 3b$ , i.e.  $b < 2$ .

*Proof.* Direct from (5.7): a row-0 cooperator has three defectors among its nine cells so  $\nu = 6$ ; a row-(-1) defector has three cooperators so  $\nu = 3$ . A row-0 cooperator's neighbourhood spans rows  $-1..1$  and hence contains row-1 cooperators of score 9; a row-(-1) defector's neighbourhood spans rows  $-2..0$  and hence contains row-0 cooperators of score 6, while the best defector score in it is  $3b$ .  $\square$

**Remark 5.19 (Why  $1 < b < 2$  is the interesting window).** The upper limit has two independent meanings that happen to coincide. Algebraically,  $b < 2$  is condition (2.2),  $2R > T + S$ , without which mutual cooperation is not the efficient outcome. Dynamically, Proposition 5.18 shows  $b < 2$  is exactly the condition for a flat cooperative front to advance into a defector region. The lower limit  $b > 1$  is  $T > R$ , without which there is no dilemma at all. Inside the window, Propositions 5.16–5.18 give a coherent picture of the two elementary processes in competition: defector clusters grow at their *corners* (needing  $b > 9/5$ ) while cooperator regions eat into them along their *flat edges* (needing  $b < 2$ ). For  $b < 9/5$  only the second process operates, defector clusters are static or shrinking, and cooperators come to occupy most of the lattice; for  $9/5 < b < 2$  both operate simultaneously, and Nowak and May's simulations from random initial conditions reveal *spatial chaos* — a persistently churning coexistence with an asymptotic cooperator frequency close to 0.3178, largely independent of  $b$  within this range and of the initial condition (Nowak & May, 1992, 1993). The intermediate thresholds  $b = 9/8$ ,  $7/5$  and  $8/5$  mark, respectively, the point above which a lone defector stops being frozen and starts to grow (for  $9/8 < b < 7/5$  the resulting  $3 \times 3$  block collapses back to a single defector and regrows, a period-two blinker), the appearance of static defector crosses, and the appearance of static defector squares, and they are exactly the values at which the observed cooperator frequency in `pdlab.spatial` changes character.

**Remark 5.20 (Robustness).** The Nowak–May result depends on modelling choices. Huberman and Glance (Huberman & Glance, 1993) showed that replacing synchronous updating by asynchronous (continuous-time) updating destroys the chaotic coexistence: defectors take over the lattice for the same  $b$ , so the persistent cooperation at large  $b$  is an artefact of the discrete-time lockstep. The modern reading is that the *mechanism* — assortment of cooperators through spatial clustering — is

robust, while the specific chaotic attractor and its 0.3178 signature are not. Both update schemes are implemented in `pdlab.spatial` and compared in Chapter 6.

**Remark 5.21 (Nowak’s five rules).** Nowak (Nowak, 2006b) organises the mechanisms that permit cooperation in the donation game  $\text{Don}(b, c)$  into five conditions, each of the form “benefit-to-cost ratio exceeds something”:

- **Kin selection:**  $r > c/b$ , where  $r$  is the coefficient of relatedness (Hamilton’s rule).
- **Direct reciprocity:**  $w > c/b$ , where  $w$  is the probability of another round — our  $\delta$ ; indeed for the donation game  $(T - R)/(T - P) = c/b$ , so Theorem 4.5 *is* this rule.
- **Indirect reciprocity:**  $q > c/b$ , where  $q$  is the probability of knowing a co-player’s reputation.
- **Network reciprocity:**  $b/c > k$ , where  $k$  is the average number of neighbours, for evolutionary graph dynamics under weak selection (Ohtsuki et al., 2006). On the Moore lattice  $k = 8$ ; the Nowak–May regime, which uses strong (imitate-the-best) selection rather than weak, is a different limit of the same idea.
- **Group selection:**  $b/c > 1 + n/m$ , where  $n$  is the group size and  $m$  the number of groups.

The first four are all present in this report in one form or another: direct reciprocity in Chapter 4, network reciprocity in Section 5.5, and the finite-population analogue of assortment in Section 5.4.

## 6 Numerical analysis: testing the theory

Every result in this chapter is produced by `pdlab`, the Python package written for this project, from a single command (`make reproduce`) with fixed random seeds. The numbers quoted in the text are substituted into the manuscript at build time directly from `results/canonical.json`, so a number in this report cannot disagree with the code that produced it. All experiments use Axelrod’s payoffs  $(T, R, P, S) = (5, 3, 1, 0)$  unless stated otherwise. Simulation parameters: 200 rounds per match and 10 repetitions for tournaments; lattices of  $100 \times 100$  sites for 300 generations; Moran populations of  $N = 50$  run for 500000 birth–death events in the mutation and noise experiments (the fixation experiment uses  $N = 4$  to 32).

### 6.1 The one-shot game and backward induction

`pdlab.finite.backward_induction` solves the  $n$ -fold repeated game by the constructive argument of Theorem 3.4: at each stage it enumerates the pure Nash equilibria of the stage game shifted by the (history-independent) continuation values. For every  $n$  it finds exactly one,  $(D, D)$ , and the resulting path is all- $D$  with row total  $nP$ :

| $n$ | SPE path all- $D$ | row total |
|-----|-------------------|-----------|
| 1   | yes               | 1         |
| 2   | yes               | 2         |
| 3   | yes               | 3         |
| 4   | yes               | 4         |
| 5   | yes               | 5         |
| 6   | yes               | 6         |
| 7   | yes               | 7         |
| 8   | yes               | 8         |
| 9   | yes               | 9         |
| 10  | yes               | 10        |

| $n$ | SPE path | all- $D$ | row total |
|-----|----------|----------|-----------|
|-----|----------|----------|-----------|

The solver refuses to proceed if a stage game has several pure equilibria (a coordination game in the test-suite triggers this), which is precisely the case in which the Benoit–Krishna (1985) cooperative equilibria of finitely repeated games become possible.

## 6.2 Discount thresholds by geometric stopping

Theorems 4.5 and 4.8 predict  $\delta_{\text{Grim}}^* = (T - R)/(T - P) = 0.5$  and  $\delta_{\text{TFT}}^* = \max\{(T - R)/(T - P), (T - R)/(R - S)\} = 0.6667$ , the second branch being the binding one for Axelrod’s payoffs. `pdlab.analytic` derives both symbolically with `sympy` (the code solves  $R/(1 - \delta) = T + \delta P/(1 - \delta)$  and  $R/(1 - \delta) = (T + \delta S)/(1 - \delta^2)$  and simplifies), and the test-suite checks that the symbolic expressions reduce to the closed forms above.

To test the *interpretation* of  $\delta$  as a continuation probability, we simulate matches whose length is geometric: after every round the match continues with probability  $\delta$ , so the expected unnormalised total is exactly  $\sum_t \delta^t \pi_t$ . We play Grim against Grim (cooperation), ALLD against Grim (permanent defection) and Suspicious TFT against TFT (which produces the alternating path  $D, C, D, C, \dots$  against a TFT opponent) for 4000 matches per value of  $\delta$  and compare the normalised mean totals  $(1 - \delta) \bar{V}$  with the exact values:

| $\delta$ | $R$ (exact) | coop. (sim)          | $(1 - \delta)T + \delta P$ | ALLD dev.<br>(sim)   | $\frac{T + \delta S}{1 + \delta}$ | alt. dev.<br>(sim)   |
|----------|-------------|----------------------|----------------------------|----------------------|-----------------------------------|----------------------|
| 0.200    | 3.000       | 3.003 $\pm$<br>0.042 | 4.200                      | 4.202 $\pm$<br>0.014 | 4.167                             | 4.170 $\pm$<br>0.027 |
| 0.400    | 3.000       | 2.932 $\pm$<br>0.057 | 3.400                      | 3.405 $\pm$<br>0.020 | 3.571                             | 3.550 $\pm$<br>0.045 |
| 0.500    | 3.000       | 2.989 $\pm$<br>0.066 | 3.000                      | 3.005 $\pm$<br>0.023 | 3.333                             | 3.319 $\pm$<br>0.051 |
| 0.600    | 3.000       | 2.980 $\pm$<br>0.072 | 2.600                      | 2.598 $\pm$<br>0.024 | 3.125                             | 3.214 $\pm$<br>0.062 |
| 0.667    | 3.000       | 2.989 $\pm$<br>0.077 | 2.333                      | 2.329 $\pm$<br>0.026 | 3.000                             | 3.053 $\pm$<br>0.065 |
| 0.800    | 3.000       | 2.931 $\pm$<br>0.082 | 1.800                      | 1.785 $\pm$<br>0.028 | 2.778                             | 2.743 $\pm$<br>0.069 |
| 0.900    | 3.000       | 3.040 $\pm$<br>0.092 | 1.400                      | 1.408 $\pm$<br>0.030 | 2.632                             | 2.615 $\pm$<br>0.074 |

Of the 21 comparisons, 2 fall outside two standard errors and 0 outside three, as expected for that many independent comparisons; the crossing points of the curves in Figure 2 fall at the predicted thresholds. Note the qualitative content: the alternating deviation is the *binding* one for TFT, which is why TFT needs a more patient player ( $\delta \geq 2/3$ ) than Grim Trigger ( $\delta \geq 1/2$ ).

## 6.3 Exact Markov-chain payoffs versus simulation

For two memory-one strategies the match is a four-state Markov chain (Chapter 7 gives the details). `pdlab.markov` computes the stationary distribution exactly, and also implements the Press–Dyson

**Cooperation vs. deviation (lines: exact; markers: simulation,  $\pm 2$  s.e.)**

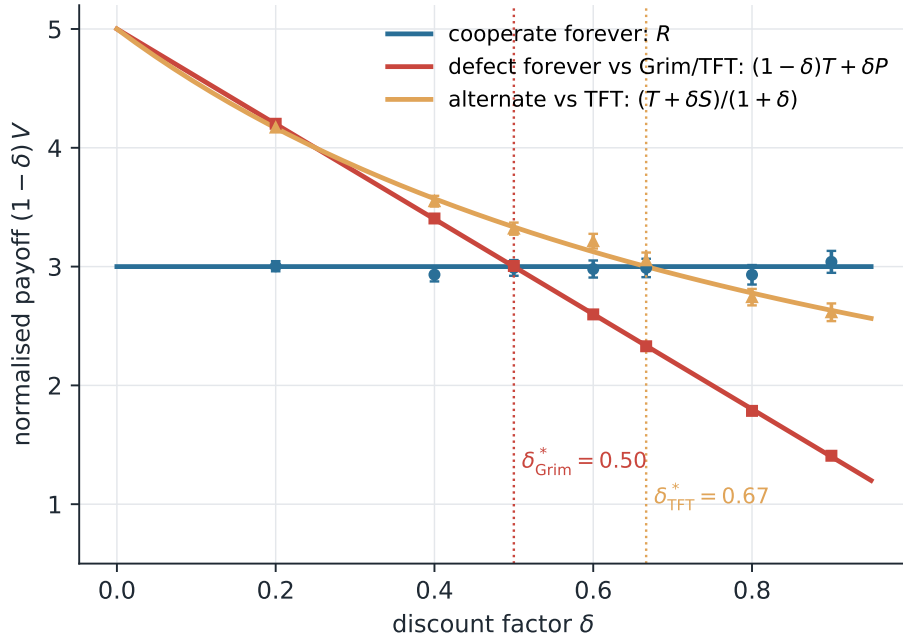


Figure 2: Cooperation and deviation values as a function of the discount factor: lines are the closed forms, markers are geometric-stopping simulations ( $\pm 2$  s.e.). The dotted lines mark  $\delta_{\text{Grim}}^* = 1/2$  and  $\delta_{\text{TFT}}^* = 2/3$ .

determinant formula  $s_X = D(p, q, S_X)/D(p, q, \mathbf{1})$  independently; the test-suite verifies that the two agree to  $10^{-9}$  on random strategy pairs and that the exact expected  $n$ -round averages agree with simulated matches for deterministic strategies. This machinery underlies every “exact” curve in the remainder of the report.

#### 6.4 An Axelrod-style tournament

We ran a round-robin among 18 strategies (the fifteen classical strategies of Table 2 plus the two zero-determinant strategies and a Q-learning agent), every strategy also playing a copy of itself, exactly as in Axelrod’s tournaments. The same tournament was then repeated with implementation noise  $\varepsilon \in \{0, 0.01, 0.05, 0.1\}$ : each intended move is replaced by its opposite with probability  $\varepsilon$ , independently for both players, and both players observe the move that was actually played.

Table 2: The tournament field. “Nice” means never the first to defect.

| strategy | rule                                                | nice | memory-one $p$           |
|----------|-----------------------------------------------------|------|--------------------------|
| ALLC     | always cooperate                                    | yes  | (1, 1, 1, 1)             |
| ALLD     | always defect                                       | no   | (0, 0, 0, 0)             |
| TFT      | cooperate first, then copy the opponent’s last move | yes  | (1, 0, 1, 0)             |
| STFT     | TFT starting with $D$                               | no   | (1, 0, 1, 0), starts $D$ |

| strategy | rule                                                                    | nice | memory-one $p$                                         |
|----------|-------------------------------------------------------------------------|------|--------------------------------------------------------|
| GTFT     | TFT that forgives a defection with probability 1/3                      | yes  | $(1, \frac{1}{3}, 1, \frac{1}{3})$                     |
| TF2T     | defect only after two consecutive defections                            | yes  | —                                                      |
| GRIM     | cooperate until the first defection, then defect forever                | yes  | —                                                      |
| WSLS     | repeat the last move after $R$ or $T$ , switch after $S$ or $P$         | yes  | $(1, 0, 0, 1)$                                         |
| CTFT     | TFT with standing (contrite)                                            | yes  | —                                                      |
| RANDOM   | cooperate with probability 1/2                                          | no   | $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ |
| ALT      | $C, D, C, D, \dots$                                                     | no   | —                                                      |
| SOFTMAJ  | cooperate iff the opponent has cooperated at least as often as defected | yes  | —                                                      |
| HARDMAJ  | cooperate iff the opponent has cooperated strictly more often           | no   | —                                                      |
| PROBER   | $D, C, C$ ; exploit if not punished, else TFT                           | no   | —                                                      |
| GRADUAL  | punish the $k$ -th defection with $k$ defections, then $C, C$           | yes  | —                                                      |
| EXTORT2  | zero-determinant extortioner, $\chi = 2$                                | no   | $(\frac{8}{9}, \frac{1}{2}, \frac{1}{3}, 0)$           |
| ZDGTFT2  | generous zero-determinant, $\chi = 2$                                   | yes  | $(1, \frac{1}{8}, 1, \frac{1}{4})$                     |
| QLEARN   | tabular $\varepsilon$ -greedy Q-learner on the last outcome             | —    | learned                                                |

Table 3: Mean payoff per round (rank in parentheses) for four noise levels. Strategies are ordered by their noiseless score.

| strategy | $\varepsilon = 0.0$ | $\varepsilon = 0.01$ | $\varepsilon = 0.05$ | $\varepsilon = 0.1$ | coop. rate<br>( $\varepsilon = 0$ ) |
|----------|---------------------|----------------------|----------------------|---------------------|-------------------------------------|
| GRADUAL  | 2.737 (1)           | 2.584 (1)            | 2.438 (1)            | 2.350 (1)           | 0.79                                |
| ZDGTFT2  | 2.665 (2)           | 2.501 (2)            | 2.318 (5)            | 2.214 (7)           | 0.86                                |

| strategy | $\varepsilon = 0.0$ | $\varepsilon = 0.01$ | $\varepsilon = 0.05$ | $\varepsilon = 0.1$ | coop. rate<br>( $\varepsilon = 0$ ) |
|----------|---------------------|----------------------|----------------------|---------------------|-------------------------------------|
| GTFT     | 2.645 (3)           | 2.459 (3)            | 2.348 (3)            | 2.255 (6)           | 0.89                                |
| CTFT     | 2.615 (4)           | 2.395 (5)            | 2.401 (2)            | 2.324 (2)           | 0.75                                |
| TFT      | 2.599 (5)           | 2.342 (7)            | 2.296 (6)            | 2.291 (3)           | 0.75                                |
| SOFTMAJ  | 2.552 (6)           | 2.410 (4)            | 2.320 (4)            | 2.261 (5)           | 0.77                                |
| TF2T     | 2.446 (7)           | 2.336 (8)            | 2.270 (8)            | 2.185 (9)           | 0.80                                |
| GRIM     | 2.401 (8)           | 1.999 (16)           | 1.959 (17)           | 2.074 (14)          | 0.56                                |
| PROBER   | 2.377 (9)           | 2.207 (10)           | 2.193 (9)            | 2.205 (8)           | 0.50                                |
| ALT      | 2.370 (10)          | 2.361 (6)            | 2.292 (7)            | 2.290 (4)           | 0.50                                |
| WSLS     | 2.368 (11)          | 2.194 (11)           | 2.085 (14)           | 2.121 (13)          | 0.76                                |
| ALLC     | 2.366 (12)          | 2.216 (9)            | 2.075 (15)           | 2.030 (18)          | 1.00                                |
| HARDMAJ  | 2.192 (13)          | 2.148 (13)           | 2.130 (11)           | 2.073 (15)          | 0.45                                |
| STFT     | 2.180 (14)          | 2.181 (12)           | 2.146 (10)           | 2.152 (10)          | 0.51                                |
| RANDOM   | 2.067 (15)          | 2.115 (14)           | 2.127 (12)           | 2.129 (12)          | 0.50                                |
| QLEARN   | 2.053 (16)          | 2.024 (15)           | 2.118 (13)           | 2.143 (11)          | 0.39                                |
| ALLD     | 1.759 (17)          | 1.819 (17)           | 1.916 (18)           | 2.052 (16)          | 0.00                                |
| EXTORT2  | 1.700 (18)          | 1.772 (18)           | 1.980 (16)           | 2.041 (17)          | 0.18                                |

Three features deserve comment.

1. **Nice strategies fill the top of the table**, exactly as Axelrod found: the 8 best strategies at  $\varepsilon = 0$  are all nice (never the first to defect); PROBER is the first that is not. Tit-for-Tat itself is fifth; the winner is GRADUAL with 2.737 per round, in line with Beaufils, Delahaye and Mathieu (1996), who introduced Gradual precisely because it beats TFT in tournaments of this kind. TFT’s success in 1980 depended on the particular field of entrants, not on dominance over all comers (indeed no strategy can be best against every field; Axelrod (1984, Chapter 2)).
2. **Generous strategies gain as noise rises**. GTFT and ZDGTFT-2 overtake the stricter reciprocators once  $\varepsilon > 0$ , while Grim collapses: a single accidental defection triggers permanent punishment. Figure 3 shows the full rank changes.
3. **Extortion loses tournaments**. Extort-2 is last without noise and never better than 16th of 18 at any noise level (only ALLD and, under heavy noise, ALLC keep it company at the bottom). It can only “win” *relative* to its current opponent, at the cost of a low absolute score. We return to this in Chapter 7.

#### 6.4.1 Cross-check against an independent implementation

To guard against implementation errors we compared `pdlab` with the independently developed `axelrod` library (Knight et al., 2016). For all 91 pairings of the thirteen deterministic strategies the two libraries produce *identical* 200-round scores, and the memory-one vectors of GTFT, Extort-2 and ZDGTFT-2 agree with the library’s to machine precision (`results/crosscheck_axelrod.md`).

### 6.5 Replicator dynamics for ALLC, ALLD and TFT

With  $\delta = 0.9$  the normalised payoff matrix of Chapter 5 is

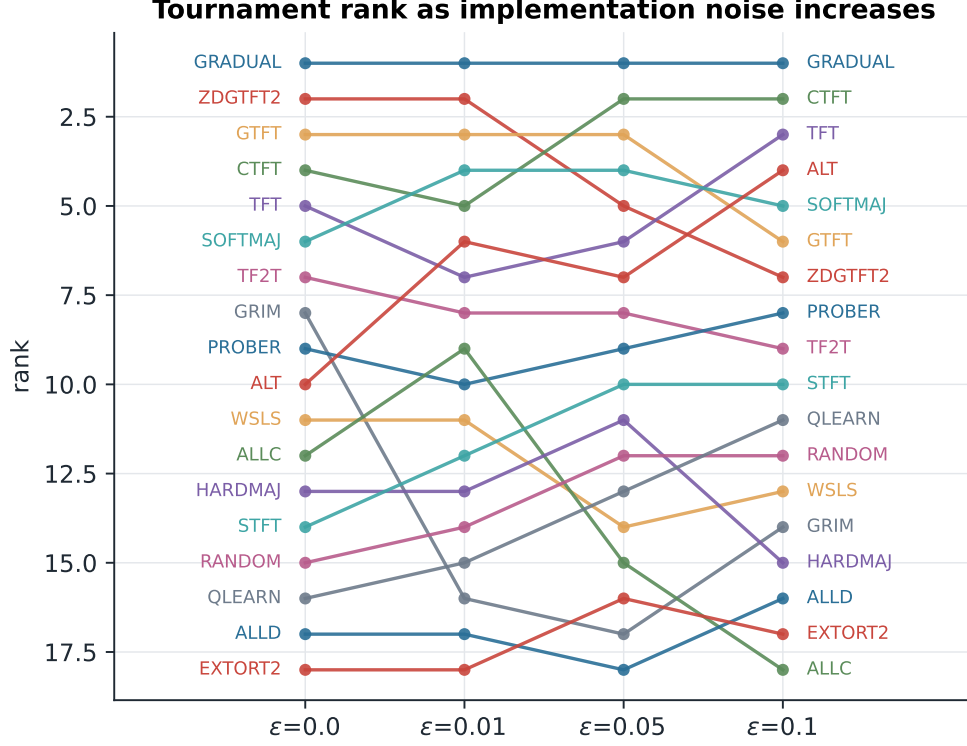


Figure 3: Tournament rank as implementation noise increases.

$$A = \begin{pmatrix} 3.00 & 0.00 & 3.00 \\ 5.00 & 1.00 & 1.40 \\ 3.00 & 0.90 & 3.00 \end{pmatrix},$$

and Theorem 5.8 predicts the unstable fixed point on the ALLD–TFT edge at  $x_{\text{TFT}}^* = 0.0588 = 1/17$ . Integrating the replicator equation (`scipy.integrate.solve_ivp`, relative tolerance  $10^{-9}$ ) from 27 initial conditions confirms the bistable picture of Figure 5: populations that start with fewer than  $x^*$  reciprocators converge to all-ALLD; populations above the threshold eliminate ALLD and drift to a point on the ALLC–TFT edge that depends on the initial condition (the edge is a continuum of neutral fixed points). The finite-difference Jacobian eigenvalues agree with the symbolic ones of Theorem 5.9:

| fixed point   | eigenvalues (symbolic)                                                                                                                                                                                                                              | numeric ( $\delta = 0.9$ ) |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| ALLC          | $0; -R + T$                                                                                                                                                                                                                                         | 2.0000, -0.0000            |
| ALLD          | $-P + S; (P - S) * (\delta - 1)$                                                                                                                                                                                                                    | -1.0000, -0.1000           |
| TFT           | $0; P * \delta - R - T * \delta + T$                                                                                                                                                                                                                | -1.6000, 0.0000            |
| TFT–ALLD–edge | $\delta * (P - S) * (P * \delta - R - T * \delta + T) / (-2 * P * \delta + P + R + S * \delta - S + T * \delta - T);$<br>$(P - S) * (\delta - 1) * (P * \delta - R - T * \delta + T) / (-2 * P * \delta + P + R + S * \delta - S + T * \delta - T)$ | 0.0941, -0.8471            |

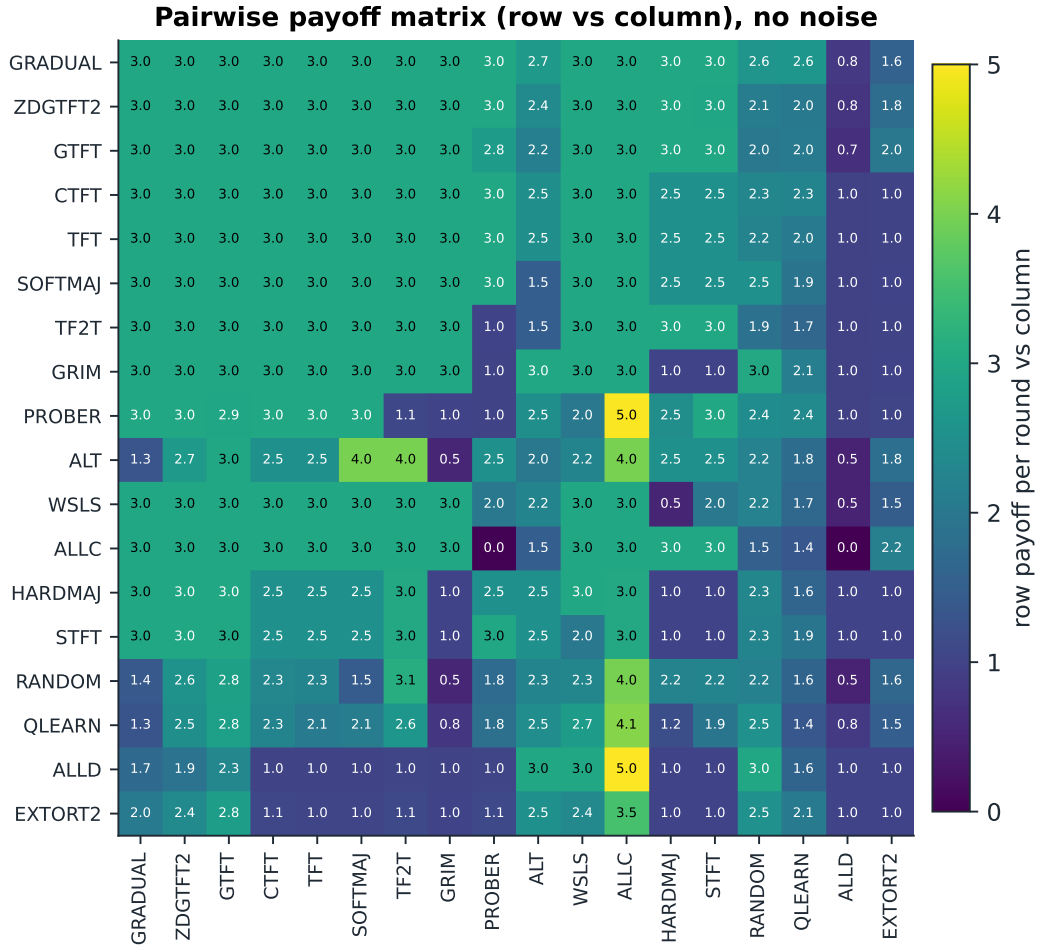


Figure 4: Pairwise per-round payoff matrix (row versus column) without noise.

### Replicator dynamics, $\delta = 0.9$ : bistability: ALLD vs. the ALLC-TFT edge

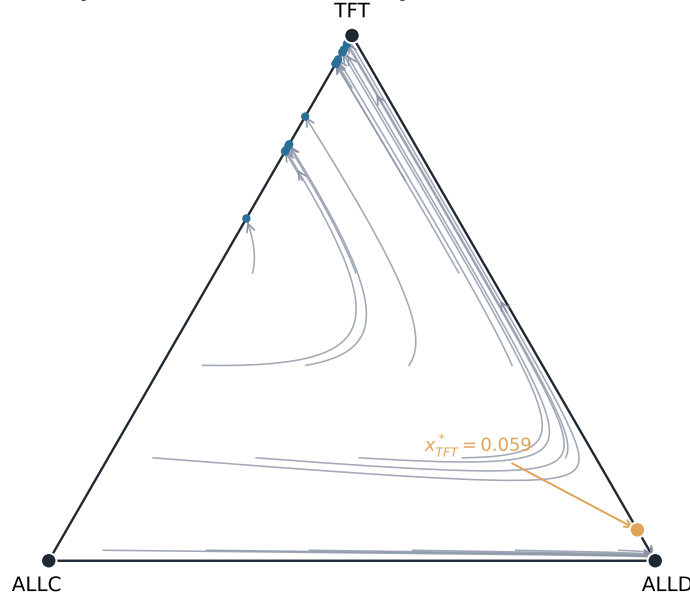


Figure 5: Replicator trajectories on the simplex for  $\delta = 0.9$ . Black dots: vertex fixed points; gold: the saddle at  $x_{\text{TFT}}^* = 1/17$ ; every point on the ALLC–TFT edge is a fixed point.

#### 6.5.1 Beyond the fixed points: how fast is convergence?

Linearisation predicts that near the ALLD vertex the TFT frequency decays as  $e^{\lambda t}$  with  $\lambda = A_{\text{TFT},\text{ALLD}} - A_{\text{ALLD},\text{ALLD}} = (1 - \delta)(S - P) = -0.100$ . A least-squares fit of  $\log x_{\text{TFT}}(t)$  on the numerical trajectory started at  $x_{\text{TFT}} = x^*/2$  gives a slope of  $-0.0984$  (Figure 6): convergence is exponential, and the rate is set by the *discounted* gap between the sucker’s and the punishment payoff. Because  $(1-\delta)$  multiplies the rate, a more patient population (larger  $\delta$ ) loses its reciprocators *more slowly*: patience protects cooperation even where it fails to sustain it.

## 6.6 Finite populations: the Moran process

In a finite population selection is stochastic. We simulate the frequency-dependent Moran process with fitness equal to payoff ( $w = 1$ ) and compare the fixation probability of a single TFT mutant in an ALLD population with the exact formula of Theorem 5.13, for 2000 independent runs per population size:

| $N$ | $\rho$ exact | $\rho$ simulated | s.e.   | $1/N$  |
|-----|--------------|------------------|--------|--------|
| 4   | 0.2589       | 0.2615           | 0.0098 | 0.2500 |
| 6   | 0.2121       | 0.2075           | 0.0091 | 0.1667 |
| 8   | 0.1867       | 0.1895           | 0.0088 | 0.1250 |
| 12  | 0.1565       | 0.1480           | 0.0079 | 0.0833 |
| 16  | 0.1371       | 0.1270           | 0.0074 | 0.0625 |
| 24  | 0.1117       | 0.1245           | 0.0074 | 0.0417 |
| 32  | 0.0949       | 0.1015           | 0.0068 | 0.0312 |

0 of the seven simulated values fall outside two standard errors of the exact ones. The fixation

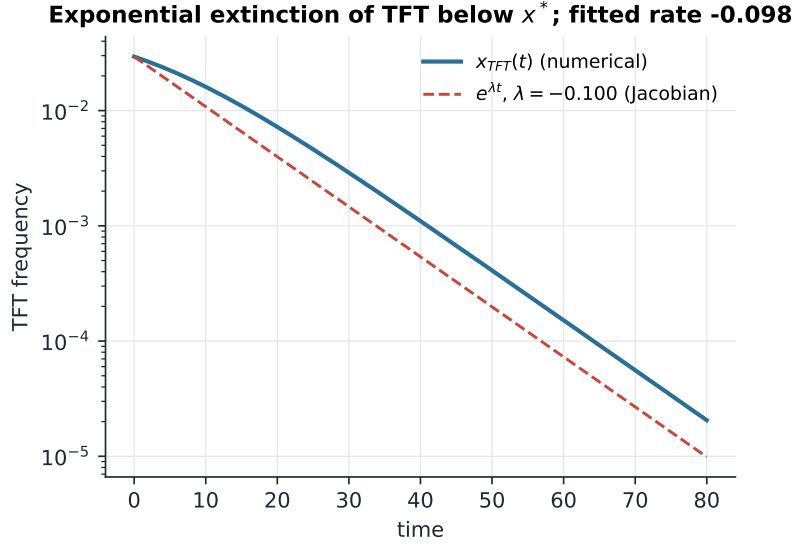


Figure 6: Extinction of TFT below the threshold is exponential at the Jacobian rate.

probability exceeds the neutral value  $1/N$  for every  $N$ : a lone reciprocator is *favoured* by selection even though TFT is not an ESS in the infinite-population sense, because with  $N$  finite it meets itself sufficiently often once it has a few copies. (The  $1/3$  law of Theorem 5.14 is a weak-selection, large- $N$  statement; here  $w = 1$  and  $N \leq 32$ , so the agreement in sign is a bonus rather than a test of that theorem.)

### 6.6.1 Mutation: how much is needed to break the cooperative equilibrium?

The syllabus asks how large the mutation rate must be to destroy the equilibrium. The answer depends sharply on the strategy space.

*Three strategies.* Starting from an all-TFT population of size  $N$  with mutation to a uniformly random strategy at rate  $\mu$  per birth:

| $\mu$ | ALLC  | ALLD  | TFT   | cooperation | time ALLD $> 1/2$ |
|-------|-------|-------|-------|-------------|-------------------|
| 0.0   | 0.000 | 0.000 | 1.000 | 1.000       | 0.00              |
| 0.001 | 0.106 | 0.104 | 0.790 | 0.813       | 0.10              |
| 0.003 | 0.083 | 0.170 | 0.747 | 0.703       | 0.17              |
| 0.01  | 0.103 | 0.192 | 0.704 | 0.673       | 0.19              |
| 0.03  | 0.146 | 0.201 | 0.653 | 0.667       | 0.18              |
| 0.1   | 0.198 | 0.243 | 0.558 | 0.621       | 0.14              |
| 0.2   | 0.223 | 0.352 | 0.426 | 0.499       | 0.22              |
| 0.3   | 0.254 | 0.374 | 0.372 | 0.487       | 0.18              |
| 0.4   | 0.277 | 0.370 | 0.353 | 0.499       | 0.13              |
| 0.5   | 0.299 | 0.359 | 0.342 | 0.518       | 0.07              |

Without mutation TFT is absorbing. With mutation the population is pushed along the neutral ALLC–TFT edge by drift; once enough ALLC has accumulated an ALLD mutant invades (the

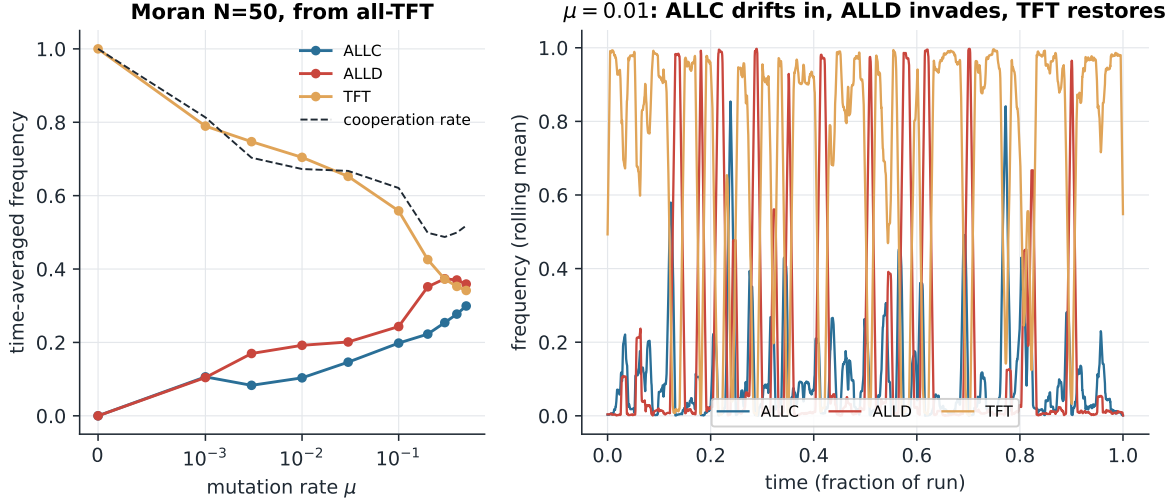


Figure 7: Mutation-driven cycling in the three-strategy Moran process.

TFT share has fallen below the threshold), the population crashes towards ALLD, and it is rescued only when a TFT lineage reaches the threshold  $x^*$  by drift. The result is the ALLC→ALLD→TFT cycle of Nowak and Sigmund (Nowak, 2006a, Chapter 5; 1992) (Figure 7). Already at  $\mu = 0.001$  (one birth in a thousand) the population spends 10% of its time with an ALLD majority and the cooperation rate has fallen from 1 to 0.81; at  $\mu = 0.01$  it is 0.67, and by  $\mu = 0.5$  the population is mutation-dominated, with all three strategies near one third and cooperation at 0.52. There is therefore no sharp critical mutation rate: the cooperative equilibrium erodes continuously, and the *first* order-of-magnitude of mutation is the one that matters, because it opens the neutral drift along the ALLC–TFT edge that ALLD then exploits.

*Eighteen strategies.* With the full tournament field and the same protocol the answer is strikingly different:

| $\mu$ | most frequent | cooperation rate | mean payoff |
|-------|---------------|------------------|-------------|
| 0.0   | SOFTMAJ       | 1.000            | 3.000       |
| 0.001 | GRIM          | 0.930            | 2.889       |
| 0.005 | GRADUAL       | 0.755            | 2.597       |
| 0.01  | GRIM          | 0.905            | 2.839       |
| 0.02  | GRIM          | 0.922            | 2.875       |
| 0.05  | CTFT          | 0.876            | 2.806       |
| 0.1   | GRADUAL       | 0.814            | 2.701       |
| 0.2   | GRADUAL       | 0.746            | 2.585       |
| 0.3   | GRADUAL       | 0.703            | 2.509       |

Cooperation *never* collapses, even at  $\mu = 0.3$  where nearly a third of all births are random. The reason is that most of the strategy space is reciprocating: a random mutant is far more likely to be a conditional cooperator than an exploiter, and conditional cooperators earn  $R$  against one another. Mutation therefore spreads the population across the “nice” cluster (which strategy happens to be most frequent is a matter of drift, hence the changing leader) rather than out of it. The lesson is

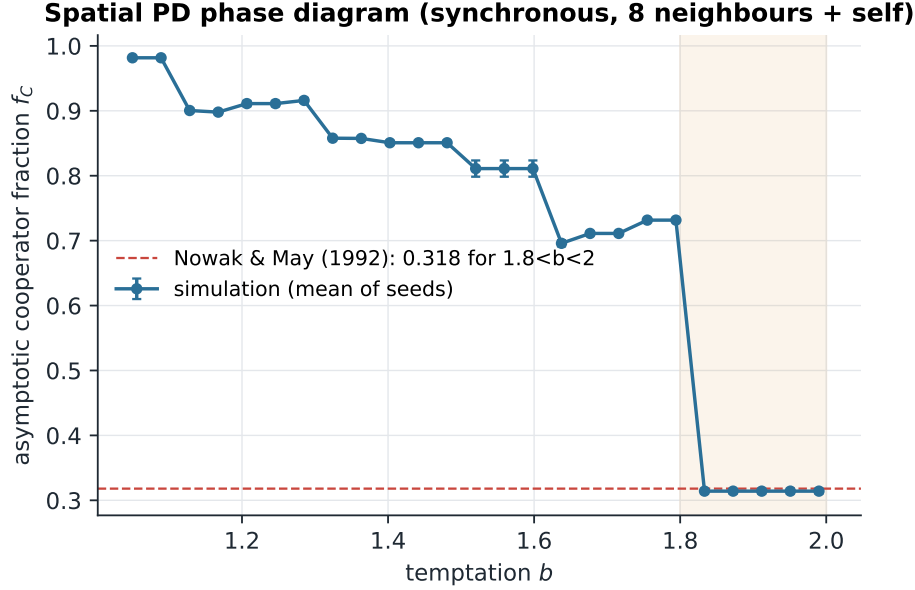


Figure 8: Asymptotic cooperator fraction versus  $b$ .

that the robustness of cooperation to mutation is a property of the *distribution of mutants*, not of the equilibrium itself.

## 6.7 Spatial structure: Nowak and May’s lattice

`pdlab.spatial` implements the weak dilemma on a square lattice with eight neighbours, self-interaction, synchronous updating and periodic boundaries. Sweeping  $b$  from 1.05 to 1.99 from random 90% cooperator initial conditions gives the phase diagram of Figure 8 (raw values in Table 4). The cooperator fraction is a staircase in  $b$ , because the outcome of every local contest depends only on the comparison of integer multiples of  $b$  against integers. The four thresholds derived in Chapter 5 ( $9/8$ ,  $7/5$ ,  $8/5$ ,  $9/5$ ) are among the step locations; the remaining steps (near 1.2, 1.32, 1.52, 1.68 and 1.76) come from other local configurations, larger clusters and irregular boundaries, that Chapter 5 does not enumerate.

For  $1.8 < b < 2$  our lattices settle into the “spatial chaos” regime with  $f_C = 0.314$  averaged over all  $b$  in that interval and over seeds, against the published value of 0.318 (Nowak & May, 1992). The 1% discrepancy is within the run-to-run fluctuation of  $f_C$  (compare the time series of Figure 9), and the value is independent of the initial cooperator fraction, as the authors reported:

| initial $f_C$ | asymptotic $f_C$ |
|---------------|------------------|
| 0.3           | 0.321            |
| 0.5           | 0.316            |
| 0.7           | 0.317            |
| 0.9           | 0.314            |
| 0.99          | 0.312            |

Starting from a single defector in a sea of cooperators with fixed boundaries reproduces Nowak and May’s kaleidoscope (Figure 10): the defector cluster grows, cooperators re-invade along the diagonals where they touch enough cooperators to out-score the defectors, and the symmetric initial condition is preserved forever by the deterministic rule.

### 6.7.1 Beyond: the update rule matters

| $b$   | $f_C$ (mean over seeds) | s.d.  |
|-------|-------------------------|-------|
| 1.167 | 0.898                   | 0.003 |
| 1.207 | 0.911                   | 0.004 |
| 1.246 | 0.911                   | 0.004 |
| 1.285 | 0.916                   | 0.004 |
| 1.324 | 0.858                   | 0.000 |
| 1.363 | 0.857                   | 0.002 |
| 1.403 | 0.851                   | 0.003 |
| 1.442 | 0.851                   | 0.003 |
| 1.481 | 0.851                   | 0.003 |
| 1.520 | 0.811                   | 0.013 |
| 1.559 | 0.811                   | 0.013 |
| 1.598 | 0.811                   | 0.013 |
| 1.637 | 0.696                   | 0.005 |
| 1.677 | 0.711                   | 0.003 |
| 1.716 | 0.711                   | 0.003 |
| 1.755 | 0.732                   | 0.001 |
| 1.794 | 0.732                   | 0.001 |
| 1.833 | 0.314                   | 0.004 |
| 1.873 | 0.314                   | 0.004 |
| 1.912 | 0.314                   | 0.004 |
| 1.951 | 0.314                   | 0.004 |
| 1.990 | 0.314                   | 0.004 |

## 7 Original contribution: noise, exact Markov analysis, and extortion against learners

The theory of Chapter 4 assumes that intended moves are executed perfectly. Real players tremble: a soldier fails to hold fire, a firm’s price cut is a clerical error, a bacterium’s toxin gene misfires. This chapter treats implementation error exactly, then turns to a second departure from the classical setting, opponents who *learn*, and asks whether Press and Dyson’s extortionate strategies really do “enslave” adaptive players.

### 7.1 Memory-one strategies as a Markov chain

Let  $p = (p_{CC}, p_{CD}, p_{DC}, p_{DD})$  and  $q$  be the cooperation probabilities of players  $X$  and  $Y$  conditional on the previous outcome (each written from the player’s own perspective, so  $Y$ ’s vector must be read in the order  $CC, DC, CD, DD$  when  $X$ ’s state labels are used). An implementation error that flips each intended move independently with probability  $\varepsilon$  replaces every probability  $p$  by

$$\tilde{p} = (1 - \varepsilon)p + \varepsilon(1 - p) = (1 - 2\varepsilon)p + \varepsilon.$$

The outcome sequence is a Markov chain on  $\{CC, CD, DC, DD\}$  with transition matrix  $M_{ss'} = \tilde{p}_s^{x'} \tilde{q}_s^{y'}$  (with the obvious complements for  $D$ ). For  $\varepsilon > 0$  every entry of  $M$  is strictly positive, so the chain is irreducible and aperiodic and has a unique stationary distribution  $v$ ; the long-run payoffs are  $s_X = v \cdot S_X$  and  $s_Y = v \cdot S_Y$  with  $S_X = (R, S, T, P)$  and  $S_Y = (R, T, S, P)$ . `pdlab.markov` com-

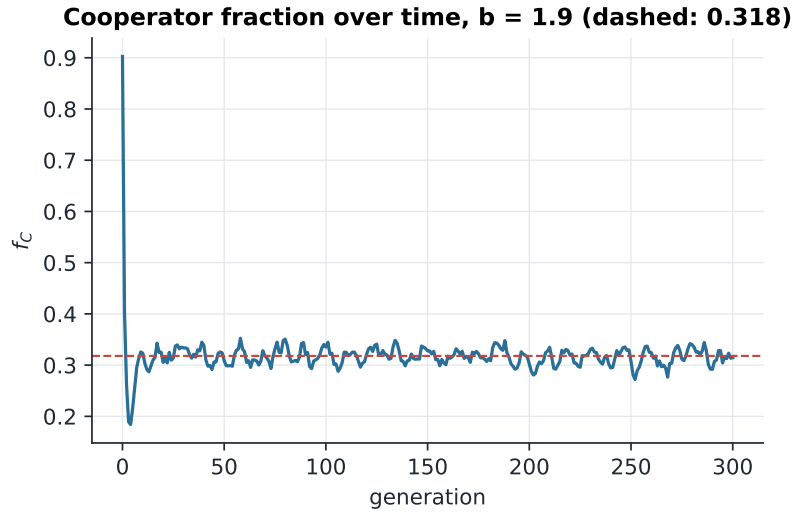


Figure 9: Cooperator fraction over time for  $b = 1.9$ ; the dashed line is 0.318.

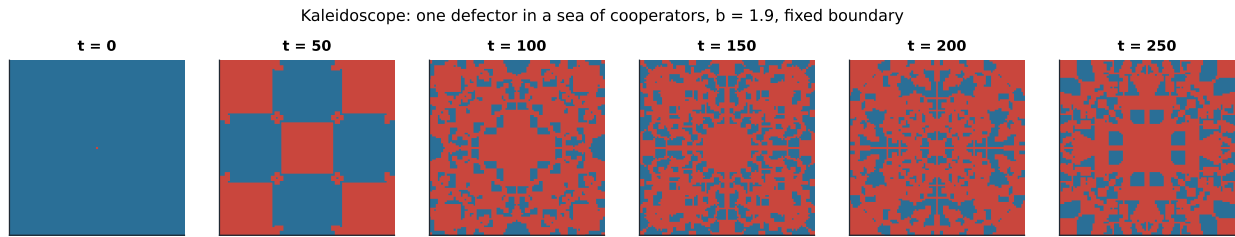


Figure 10: Deterministic “evolutionary kaleidoscope” from a single defector,  $b = 1.9$ , fixed boundary.

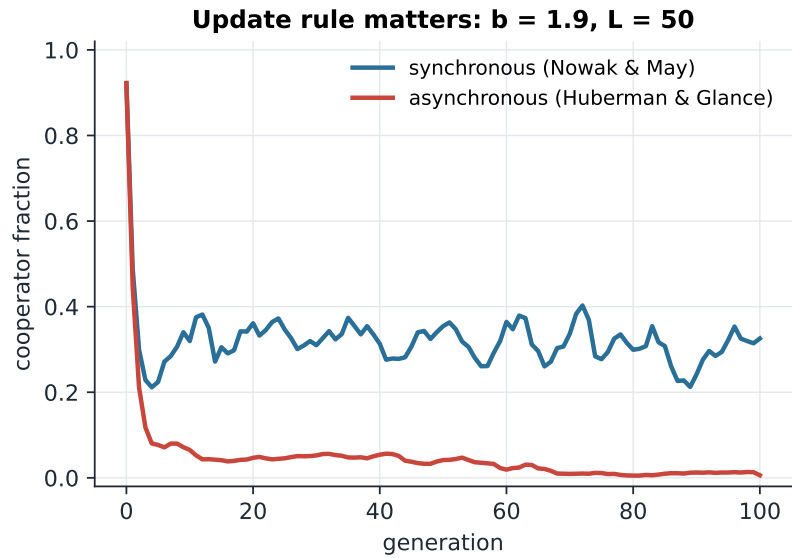


Figure 11: Synchronous versus asynchronous updating from the same initial condition,  $b = 1.9$ .

putes  $v$  numerically; `pdlab.analytic.stationary_payoffs_symbolic` computes it *symbolically* in  $(T, R, P, S, \varepsilon)$  with `sympy`.

**Proposition 7.1 (Exact self-play payoffs under noise).** Let  $(T, R, P, S)$  be a strict PD and  $\varepsilon \in (0, \frac{1}{2}]$ . The self-play payoff per round of

1. TFT is  $\frac{T + R + P + S}{4}$ , independent of  $\varepsilon$ ;
2. WSLS is  $R - (4R - 2P - T - S)\varepsilon + O(\varepsilon^2)$ ;
3. ALLC is  $R - (2R - T - S)\varepsilon + O(\varepsilon^2)$  and ALLD is  $P + (T + S - 2P)\varepsilon + O(\varepsilon^2)$  (for a pair of unconditional players the four outcomes have probabilities  $(1 - \varepsilon)^2$ ,  $\varepsilon(1 - \varepsilon)$ ,  $\varepsilon(1 - \varepsilon)$ ,  $\varepsilon^2$ , which gives both expansions directly);
4. GTFT with forgiveness  $g = 1/3$  is, for  $(5, 3, 1, 0)$ ,  $\frac{3 + 21\varepsilon + 27\varepsilon^2}{(1 + 4\varepsilon)^2} = \frac{3(1 + 7\varepsilon + 9\varepsilon^2)}{(1 + 4\varepsilon)^2} = 3 - 3\varepsilon + 3\varepsilon^2 + O(\varepsilon^3)$ .

*Proof.* (1) Write  $a = 1 - \varepsilon$ . In  $X$ 's state order  $(CC, CD, DC, DD)$  the noisy TFT rule gives  $X$  the cooperation probabilities  $\tilde{p} = (a, \varepsilon, a, \varepsilon)$  (cooperate iff the opponent cooperated) and gives  $Y$  the probabilities  $\tilde{q} = (a, a, \varepsilon, \varepsilon)$  (cooperate iff  $X$  cooperated). As  $s$  runs over the four states the pair  $(\tilde{p}_s, \tilde{q}_s)$  runs over all four elements of  $\{a, \varepsilon\}^2$  exactly once. Hence for every target state  $s'$  the column sum  $\sum_s M_{ss'} = \sum_s \tilde{p}_s^{x'} (1 - \tilde{p}_s)^{1-x'} \tilde{q}_s^{y'} (1 - \tilde{q}_s)^{1-y'}$  factorises as  $(a + \varepsilon)(a + \varepsilon) = 1$ : the chain is doubly stochastic, so the uniform vector is stationary, and since the chain is irreducible for  $\varepsilon > 0$  it is the unique stationary distribution. Therefore  $s_X = \frac{1}{4}(R + S + T + P)$ . Items (2)–(4) are direct symbolic computation (`self_play_noise_symbolic` in `pdlab.analytic` solves  $vM = v$  with `sympy`; the exact rational functions are stored in `results/canonical.json` and checked by the test-suite), followed by a Taylor expansion at  $\varepsilon = 0$ .  $\square$

Item (1) is the sharpest possible statement of TFT's fragility: the limit  $\varepsilon \rightarrow 0$  of the noisy payoff,  $9/4$ , is *not* the noiseless payoff  $R = 3$ . TFT self-play is discontinuous in the noise level. Every other cooperative strategy degrades linearly, with a slope that measures how much damage one error does before the pair recovers: ALLC ( $-1$ , unconditional, so an error costs one round), GTFT ( $-3$ ), WSLS ( $-5$ ), ZDGTFT-2 ( $-8$ ) for Axelrod's payoffs. The slope ranks self-play robustness only; who actually survives selection is the next question. Table 5 lists the expansions and the exact values at two noise levels; Figure 12 compares the exact curves with simulated matches of 20000 rounds.

Table 5: Exact self-play payoffs under implementation error.

| strategy | $p$                    | self-play payoff,                     |                         |                      |
|----------|------------------------|---------------------------------------|-------------------------|----------------------|
|          |                        | $\varepsilon \rightarrow 0$ expansion | $\varepsilon = 10^{-9}$ | $\varepsilon = 0.25$ |
| TFT      | (1, 0, 1, 0)           | $9/4$                                 | 2.250                   | 2.250                |
| GTFT     | (1, 0.333, 1, 0.333)   | $3 - 3\varepsilon$                    | 3.000                   | 2.484                |
| WSLS     | (1, 0, 0, 1)           | $3 - 5\varepsilon$                    | 3.000                   | 2.312                |
| STFT     | (1, 0, 1, 0)           | $9/4$                                 | 2.250                   | 2.250                |
| ALLC     | (1, 1, 1, 1)           | $3 - \varepsilon$                     | 3.000                   | 2.687                |
| ALLD     | (0, 0, 0, 0)           | $3\varepsilon + 1$                    | 1.000                   | 1.687                |
| EXTORT2  | (0.889, 0.5, 0.333, 0) | $18\varepsilon + 1$                   | 1.000                   | 2.098                |

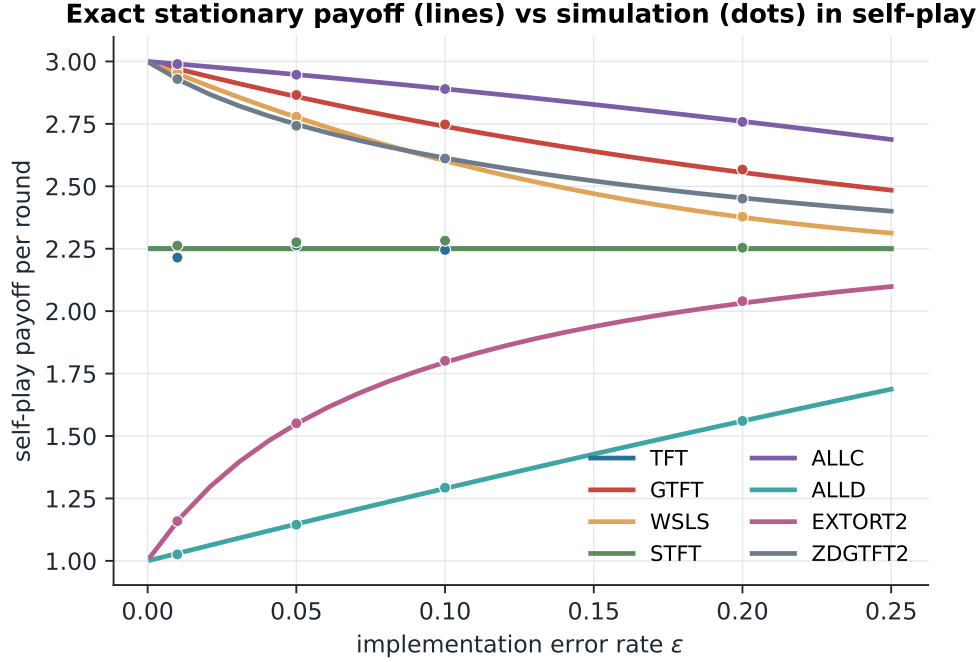


Figure 12: Self-play payoff versus error rate: lines are exact stationary payoffs, dots are simulations.

| strategy | $p$                 | self-play payoff,                     |                         |                      |
|----------|---------------------|---------------------------------------|-------------------------|----------------------|
|          |                     | $\varepsilon \rightarrow 0$ expansion | $\varepsilon = 10^{-9}$ | $\varepsilon = 0.25$ |
| ZDGTFT2  | (1, 0.125, 1, 0.25) | $3 - 8\varepsilon$                    | 3.000                   | 2.400                |

### 7.1.1 Which strategy survives selection under noise?

Self-play payoffs decide who wins once a strategy has taken over, but not who takes over. To answer that we ran the full 18-strategy tournament at seven noise levels, used each resulting payoff matrix as the fitness landscape of a Moran process ( $N = 50$ ,  $\mu = 0.01$ , 500000 events) and recorded the time-averaged frequencies:

Table 6: Strategies favoured by selection as implementation error rises.

| $\varepsilon$ | top three (time-averaged frequency)         | mean payoff |
|---------------|---------------------------------------------|-------------|
| 0.0           | TFT (0.32), GRIM (0.18), GTFT (0.15)        | 2.926       |
| 0.005         | ZDGTFT2 (0.23), CTFT (0.20), GTFT (0.15)    | 2.816       |
| 0.01          | GRADUAL (0.37), GTFT (0.23), ZDGTFT2 (0.12) | 2.785       |
| 0.02          | GRIM (0.18), ZDGTFT2 (0.17), WSLS (0.13)    | 2.272       |
| 0.05          | CTFT (0.30), SOFTMAJ (0.15), GTFT (0.11)    | 2.554       |
| 0.1           | CTFT (0.34), SOFTMAJ (0.21), GTFT (0.13)    | 2.494       |
| 0.15          | CTFT (0.27), SOFTMAJ (0.13), GRADUAL (0.12) | 2.233       |

Without noise the strict reciprocators (TFT, Grim) hold the population. For small positive error

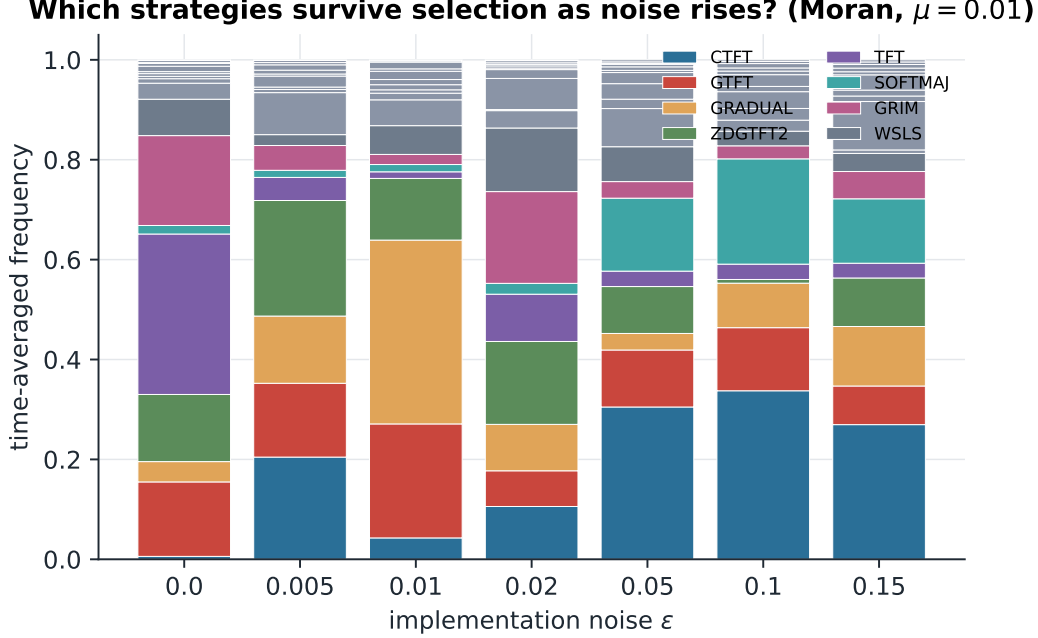


Figure 13: Time-averaged strategy frequencies in the Moran process at increasing noise.

rates the leader is decided by drift among the forgiving strategies (generous ZD, GTFT, Gradual, and at  $\varepsilon = 0.02$  even Grim, which a single long-lived lineage can carry through one run), a consequence of their near-equal payoffs against one another in a population of  $N = 50$  with  $\mu = 0.01$ . From  $\varepsilon = 0.05$  upwards the picture is unambiguous: *contrite* TFT dominates at every noise level, with the history-averaging majority rule SOFTMAJ second. CTFT's advantage is exactly the mechanism of Chapter 4: it accepts one round of punishment for its own error instead of echoing it, so its self-play payoff stays near  $R$  while TFT's collapses to  $(T + R + P + S)/4$ . (An earlier version of the code cleared contrition one round too early, which delayed rather than broke the echo; with that bug CTFT fell to mid-table under noise and the generous strategies led. The test-suite now pins the correct standing dynamics.) This is the evolutionary counterpart of Proposition 7.1: selection under noise rewards error-correction first and generosity second (Figure 13).

## 7.2 Zero-determinant strategies

Press and Dyson (2012) discovered that a memory-one player can *unilaterally* enforce a linear relation between the two long-run payoffs. Writing  $\tilde{p} = (p_{CC} - 1, p_{CD} - 1, p_{DC}, p_{DD})$ , if

$$\tilde{p} = \alpha S_X + \beta S_Y + \gamma \mathbf{1}$$

then  $\alpha s_X + \beta s_Y + \gamma = 0$  for *every* opponent  $q$  (the proof uses the determinant formula  $s = D(p, q, S)/D(p, q, \mathbf{1})$ , in which the column  $\tilde{p}$  appears; `pdlab.markov.press_dyson_payoffs` implements it and the test-suite verifies it against the stationary distribution to  $10^{-9}$ ). Two families matter here.

*Extortion.* Choosing  $\alpha = \phi$ ,  $\beta = -\phi\chi$ ,  $\gamma = \phi(\chi - 1)P$  with  $\chi > 1$  enforces  $s_X - P = \chi(s_Y - P)$ : the extortioner's surplus over mutual defection is always  $\chi$  times the opponent's. For  $(5, 3, 1, 0)$ ,  $\chi = 2$  and  $\phi = 1/18$  this is Extort-2 =  $(0.8889, 0.5, 0.3333, 0)$ , the vector published by Stewart and

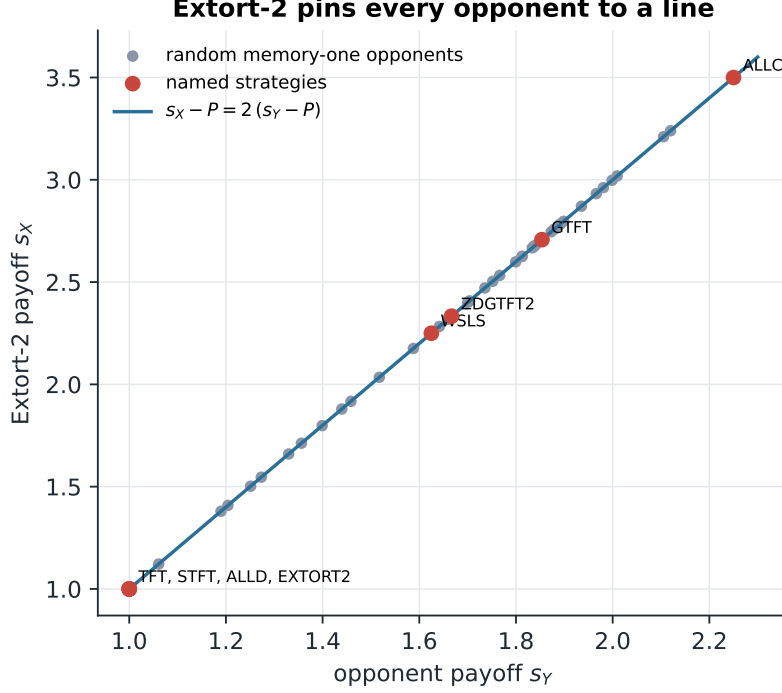


Figure 14: Without noise, Extort-2 confines every opponent to the line  $s_X - 1 = 2(s_Y - 1)$  (grey: random memory-one opponents; red: the named strategies).

Plotkin (2012) and identical to the one in the `axelrod` library. The exact noise-free payoffs against every memory-one strategy of the tournament field confirm the relation wherever the opponent earns more than  $P$  (against ALLD and against itself both players earn exactly  $P$ , so the relation holds as  $0 = 0$ ). The last column shows what happens when the *extortioneer* itself trembles: noise on  $p$  destroys the zero-determinant structure, and the enforced ratio drifts away from  $\chi = 2$ , most for opponents that punish (TFT) and least for the unconditional ones. Extortion is exact only for a player who never misplays.

Table 7: Extort-2 against named strategies: exact long-run payoffs without noise, and the enforced ratio with and without implementation error on both sides.

| opponent | $s_Y$ | $s_X$ (Extort-2) | $(s_X - P)/(s_Y - P),$<br>$\varepsilon = 0$ | same, $\varepsilon = 0.01$ |
|----------|-------|------------------|---------------------------------------------|----------------------------|
| TFT      | 1.000 | 1.000            | —                                           | 1.214                      |
| GTFT     | 1.854 | 2.707            | 2.000                                       | 1.993                      |
| WLS      | 1.625 | 2.250            | 2.000                                       | 1.884                      |
| STFT     | 1.000 | 1.000            | —                                           | 1.214                      |
| ALLC     | 2.250 | 3.500            | 2.000                                       | 2.070                      |
| ALLD     | 1.000 | 1.000            | —                                           | 0.146                      |
| EXTORT2  | 1.000 | 1.000            | —                                           | 1.000                      |
| ZDGTFT2  | 1.667 | 2.333            | 2.000                                       | 1.937                      |

*Generosity.* The same construction with baseline  $R$  instead of  $P$  enforces  $s_X - R = \chi(s_Y - R)$ :

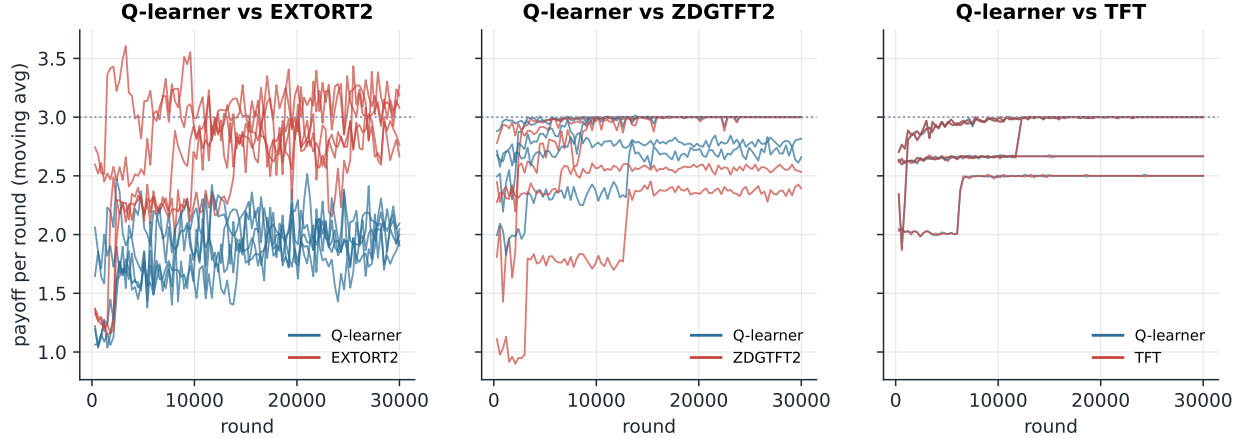


Figure 15: Payoffs per round of the learner and its opponent (moving averages), 5 seeds per panel.

the generous player’s *shortfall* from  $R$  is  $\chi$  times the opponent’s, so it always earns *less* than its opponent but pulls both towards  $R$ . With  $\chi = 2$  and the largest feasible  $\phi$  this is ZDGTFT-2 = (1, 0.125, 1, 0.25) (Stewart & Plotkin, 2012).

### 7.2.1 Extortion against an adaptive opponent

Press and Dyson’s most provocative claim is that against an *evolutionary* opponent, one that adjusts its own strategy to increase its own payoff, the extortioner obtains the maximum of its enforced line, because the opponent’s best response is to cooperate fully. We test this with a tabular Q-learner whose state is the last joint outcome (learning rate 0.05, discount 0.9, exploration probability decaying from 0.2 by a factor 0.9997 per round; these are the learner’s own parameters, unrelated to the ZD coefficients and the error rate above), playing 30000 rounds against Extort-2, ZDGTFT-2 and TFT for 5 seeds each. Table 8 reports the mean payoffs over the final fifth of each match.

Table 8: A Q-learner against zero-determinant and reciprocal opponents (final 20% of 30000 rounds).

| opponent | seeds | learner payoff<br>(mean) | opponent payoff<br>(mean) | learner range |
|----------|-------|--------------------------|---------------------------|---------------|
| EXTORT2  | 5     | 1.970                    | 2.978                     | 1.88–2.10     |
| ZDGTFT2  | 5     | 2.896                    | 2.786                     | 2.70–3.00     |
| TFT      | 5     | 2.833                    | 2.833                     | 2.50–3.00     |

Three conclusions. First, the learner is indeed exploited: against Extort-2 it ends near 2 per round while the extortioner collects about 3, and every terminal pair lies on the extortion line to within simulation error. Second, the learner does *not* reach the full cooperation that would maximise its own payoff ( $s_Y = 2.25$  against Extort-2), because a memory-one Q-learner with a discount factor of 0.9 sees the immediate temptation payoff clearly and the delayed cost only dimly; Press and Dyson’s argument assumes an opponent that optimises its *long-run* score, and the gap between that idealisation and an actual learning rule is worth about 0.25 per round. Third, generosity is rewarded: against ZDGTFT-2 and TFT the learner discovers mutual cooperation in most runs

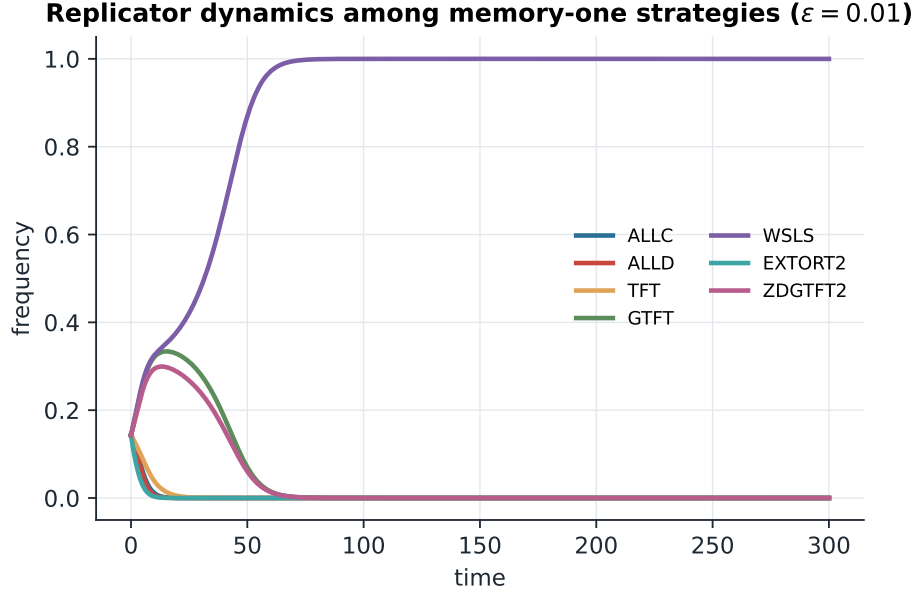


Figure 16: Replicator trajectories among memory-one strategies at  $\varepsilon = 0.01$ .

(payoff 3 for both), and where it does not it falls into a suboptimal alternating pattern rather than into exploitation.

### 7.2.2 Extortion in a population

Extortion never earns less than an opponent who earns more than  $P$  (the enforced relation gives  $s_X - P = 2(s_Y - P) \geq s_Y - P$ ), but, as the tournament of Chapter 6 already showed, it loses the field. Its evolutionary fate among memory-one strategies is decided by self-play: Extort-2 against itself earns only  $P + 18\varepsilon + O(\varepsilon^2)$  (Table 5), essentially mutual defection, so it cannot hold a population it has conquered. Replicator dynamics among seven memory-one strategies with exact payoffs at  $\varepsilon = 0.01$ , and a Moran process with mutation on the same matrix, give:

Table 9: Evolution among memory-one strategies at  $\varepsilon = 0.01$ .

| strategy | replicator, $t = 300$ | Moran ( $\mu = 0.01$ ), time average |
|----------|-----------------------|--------------------------------------|
| ALLC     | 0.000                 | 0.036                                |
| ALLD     | 0.000                 | 0.178                                |
| TFT      | 0.000                 | 0.047                                |
| GTFT     | 0.000                 | 0.094                                |
| WSLS     | 1.000                 | 0.307                                |
| EXTORT2  | 0.000                 | 0.027                                |
| ZDGTFT2  | 0.000                 | 0.310                                |

The deterministic dynamics converge to WSLS, the error-correcting strategy of Nowak and Sigmund (1993); the stochastic process with mutation spends most of its time in ZDGTFT-2 and WSLS. Extort-2 is driven to extinction by the replicator dynamics and to a mutation–selection residue of

a few per cent in the Moran process, in agreement with Hilbe, Nowak and Sigmund (2013) and Adami and Hintze (2013): extortion can act as a catalyst for cooperation but is never its end state.

### 7.3 Summary of the contribution

The exact Markov-chain treatment turns “TFT is fragile under noise” into a theorem with a striking form (a payoff discontinuous at  $\varepsilon = 0$ ) and ranks the classical repairs by the slope of their payoff in  $\varepsilon$ ; the evolutionary experiment shows selection acting on exactly that ranking. The learning experiment confirms Press and Dyson’s extortion result qualitatively while quantifying the gap between an idealised adaptive opponent and a real learning rule. All of it is reproducible from `make reproduce`.

## 8 Conclusion

The Prisoner’s Dilemma is one game with three very different theories attached to it, and this report has tried to show how they fit together by treating each of them both analytically and numerically.

*Rationality.* In the one-shot game defection strictly dominates and  $(D, D)$  is the unique Nash equilibrium (Theorem 2.8); finite repetition changes nothing (Theorems 3.4 and 3.5); infinite repetition with discount factor  $\delta$  changes everything, but only in the sense of the folk theorems: cooperation becomes *sustainable*, by Grim Trigger for  $\delta \geq (T - R)/(T - P)$  and by Tit-for-Tat as a Nash equilibrium for  $\delta \geq \max\{(T - R)/(T - P), (T - R)/(R - S)\}$ , while TFT is subgame perfect only on the knife-edge  $T - R = P - S$  (Theorem 4.9). Simulated matches with geometric stopping reproduce every threshold.

*Selection.* The replicator dynamics of  $\{\text{ALLC}, \text{ALLD}, \text{TFT}\}$  is bistable: ALLD is the unique ESS, TFT is only neutrally stable, and a population needs a reciprocator share above  $x^* = 0.0588$  (at  $\delta = 0.9$ ) to escape defection. The numerical trajectories, the symbolic Jacobian spectra and the fitted convergence rate all agree. In finite populations a single reciprocator is nevertheless *favoured* by selection, with fixation probability above  $1/N$  and matching Nowak’s exact formula. Mutation erodes cooperation in the three-strategy world through the ALLC $\rightarrow$ ALLD $\rightarrow$ TFT cycle but not in a rich strategy space, where random mutants are mostly conditional cooperators. On a lattice, cooperation survives without any memory at all, with  $f_C = 0.314$  against the published 0.318, but the survival at large  $b$  is hostage to synchronous updating.

*Noise and adaptation.* Treating memory-one play as a Markov chain gives exact payoffs under implementation error: TFT’s self-play payoff drops discontinuously to  $(T + R + P + S)/4$  for any  $\varepsilon > 0$ , whereas the generous strategies lose only  $O(\varepsilon)$ , and selection under noise tracks exactly that ranking. Zero-determinant extortion pins every opponent, including a Q-learner, to a line on which the extortioner earns more, yet it loses tournaments and goes extinct in populations, because it cannot cooperate with itself, and is driven to extinction or to a small mutation–selection residue.

The through-line is that *which theory applies is a modelling decision*: the same game predicts defection, cooperation, or coexistence depending on whether players reason, are selected, tremble, or learn. The laboratory built for this project makes those decisions explicit parameters, and every claim above can be regenerated from one command.

### 8.1 Reproducibility and verification

The accompanying repository contains:

- `src/pdlab/`: a typed Python package (game, strategies, exact Markov analysis, symbolic thresholds, replicator and Moran dynamics, spatial lattice, Q-learning, tournament engine, experiments, figures, CLI).
- `tests/`: about 350 tests including property-based tests (Hypothesis) of the payoff invariants, exact checks of every closed form against simulation, and symbolic-versus-numerical checks of the Jacobian spectra; coverage is 99% and continuous integration runs the suite, `ruff` and `mypy` on every push.
- `results/canonical.json`: every number in this report, regenerated by `make reproduce` in about two minutes and substituted into the manuscript at build time.
- `results/crosscheck_axelrod.md`: agreement with the independent `axelrod` library on all 91 deterministic pairings and on all three memory-one vectors.
- `app/index.html`: an interactive, self-contained version of the experiments in which every displayed constant is taken from `canonical.json`.

Three published values were cross-checked: the Nowak–May cooperator fraction (0.318; ours 0.314), the Stewart–Plotkin zero-determinant vectors (exact agreement), and the Beaufils–Delahaye–Mathieu result that Gradual outranks TFT in round-robin play (reproduced).

Adami, C., & Hintze, A. (2013). Evolutionary instability of zero-determinant strategies demonstrates that winning is not everything. *Nature Communications*, 4, 2193. <https://doi.org/10.1038/ncomms3193>

Axelrod, R. (1984). *The evolution of cooperation*. Basic Books.

Axelrod, R., & Hamilton, W. D. (1981). The evolution of cooperation. *Science*, 211(4489), 1390–1396. <https://doi.org/10.1126/science.7466396>

Beaufils, B., Delahaye, J.-P., & Mathieu, P. (1996). Our meeting with gradual: A good strategy for the iterated prisoner’s dilemma. In C. G. Langton & K. Shimohara (Eds), *Artificial life v: Proceedings of the fifth international workshop on the synthesis and simulation of living systems*. MIT Press.

Benoit, J.-P., & Krishna, V. (1985). Finitely repeated games. *Econometrica*, 53(4), 905–922.

Boerlijst, M. C., Nowak, M. A., & Sigmund, K. (1997). The logic of contrition. *Journal of Theoretical Biology*, 185(3), 281–293. <https://doi.org/10.1006/jtbi.1996.0326>

Flood, M. M. (1958). Some experimental games. *Management Science*, 5(1), 5–26. <https://doi.org/10.1287/mnsc.5.1.5>

Friedman, J. W. (1971). A non-cooperative equilibrium for supergames. *The Review of Economic Studies*, 38(1), 1–12.

Fudenberg, D., & Maskin, E. (1986). The folk theorem in repeated games with discounting or with incomplete information. *Econometrica*, 54(3), 533–554.

Fudenberg, D., & Tirole, J. (1991). *Game theory*. MIT Press.

Hilbe, C., Nowak, M. A., & Sigmund, K. (2013). Evolution of extortion in iterated Prisoner’s Dilemma games. *Proceedings of the National Academy of Sciences*, 110(17), 6913–6918. <https://doi.org/10.1073/pnas.1214834110>

Hofbauer, J., & Sigmund, K. (1998). *Evolutionary games and population dynamics*. Cambridge University Press. <https://doi.org/10.1017/CBO9781139173179>

Huberman, B. A., & Glance, N. S. (1993). Evolutionary games and computer simulations. *Proceedings of the National Academy of Sciences*, 90(16), 7716–7718. <https://doi.org/10.1073/pnas.90.16.7716>

Knight, V., Campbell, O., Harper, M., Langner, K., Campbell, J., Campbell, T., Carney, A., Chorley, M., Davidson-Pilon, C., Glass, K., Glynatsi, N., Ehrlich, T., Jones, M., Koutsovoulos, G., Tibble, H., Müller, J., Palmer, G., Petunov, P., Slavin, P., ... Molden, K. (2016). An

- open framework for the reproducible study of the iterated prisoner's dilemma. *Journal of Open Research Software*, 4(1), e35. <https://doi.org/10.5334/jors.125>
- Kreps, D. M., Milgrom, P., Roberts, J., & Wilson, R. (1982). Rational cooperation in the finitely repeated prisoners' dilemma. *Journal of Economic Theory*, 27(2), 245–252. [https://doi.org/10.1016/0022-0531\(82\)90029-1](https://doi.org/10.1016/0022-0531(82)90029-1)
- Maynard Smith, J., & Price, G. R. (1973). The logic of animal conflict. *Nature*, 246(5427), 15–18. <https://doi.org/10.1038/246015a0>
- Molander, P. (1985). The optimal level of generosity in a selfish, uncertain environment. *Journal of Conflict Resolution*, 29(4), 611–618.
- Nowak, M. A. (2006a). *Evolutionary dynamics: Exploring the equations of life*. Belknap Press of Harvard University Press.
- Nowak, M. A. (2006b). Five rules for the evolution of cooperation. *Science*, 314(5805), 1560–1563. <https://doi.org/10.1126/science.1133755>
- Nowak, M. A., Bonhoeffer, S., & May, R. M. (1994). Spatial games and the maintenance of cooperation. *Proceedings of the National Academy of Sciences*, 91(11), 4877–4881. <https://doi.org/10.1073/pnas.91.11.4877>
- Nowak, M. A., & May, R. M. (1992). Evolutionary games and spatial chaos. *Nature*, 359(6398), 826–829. <https://doi.org/10.1038/359826a0>
- Nowak, M. A., & May, R. M. (1993). The spatial dilemmas of evolution. *International Journal of Bifurcation and Chaos*, 3(1), 35–78. <https://doi.org/10.1142/S0218127493000040>
- Nowak, M. A., Sasaki, A., Taylor, C., & Fudenberg, D. (2004). Emergence of cooperation and evolutionary stability in finite populations. *Nature*, 428(6983), 646–650. <https://doi.org/10.1038/nature02414>
- Nowak, M. A., & Sigmund, K. (1992). Tit for tat in heterogeneous populations. *Nature*, 355(6357), 250–253. <https://doi.org/10.1038/355250a0>
- Nowak, M. A., & Sigmund, K. (1993). A strategy of win-stay, lose-shift that outperforms tit-for-tat in the Prisoner's Dilemma game. *Nature*, 364(6432), 56–58. <https://doi.org/10.1038/364056a0>
- Ohtsuki, H., Hauert, C., Lieberman, E., & Nowak, M. A. (2006). A simple rule for the evolution of cooperation on graphs and social networks. *Nature*, 441(7092), 502–505. <https://doi.org/10.1038/nature04605>
- Osborne, M. J., & Rubinstein, A. (1994). *A course in game theory*. MIT Press.
- Press, W. H., & Dyson, F. J. (2012). Iterated Prisoner's Dilemma contains strategies that dominate any evolutionary opponent. *Proceedings of the National Academy of Sciences*, 109(26), 10409–10413. <https://doi.org/10.1073/pnas.1206569109>
- Selten, R., & Stoecker, R. (1986). End behavior in sequences of finite Prisoner's Dilemma supergames: A learning theory approach. *Journal of Economic Behavior & Organization*, 7(1), 47–70. [https://doi.org/10.1016/0167-2681\(86\)90021-1](https://doi.org/10.1016/0167-2681(86)90021-1)
- Stewart, A. J., & Plotkin, J. B. (2012). Extortion and cooperation in the Prisoner's Dilemma. *Proceedings of the National Academy of Sciences*, 109(26), 10134–10135. <https://doi.org/10.1073/pnas.1208087109>
- Taylor, P. D., & Jonker, L. B. (1978). Evolutionary stable strategies and game dynamics. *Mathematical Biosciences*, 40(1–2), 145–156. [https://doi.org/10.1016/0025-5564\(78\)90077-9](https://doi.org/10.1016/0025-5564(78)90077-9)
- Watkins, C. J. C. H., & Dayan, P. (1992). Q-learning. *Machine Learning*, 8(3–4), 279–292. <https://doi.org/10.1007/BF00992698>